



THE UNIVERSITY OF
MELBOURNE

Probability Reliability and Quality

Reliability Estimation

Kevin Otto

Schedule and Learning Outcomes

Week	Recorded Lecture Topic
1	Introduction.
2	Inspection. Yield.
3	ANOVA. Measurement Systems Analysis.
4	Statistics. Confidence Intervals.
5	Hypothesis Testing. Sample Sizes.
6	Quality Management Processes.
7	Regression.
8	Design of Experiments.
9	System Reliability. Availability.
	<i>BREAK – Non Teaching Week</i>
10	Parametric Distribution Analysis
11	Statistical Process Control. Supplier Quality.
12	Review

After completion of the course the student should be able to:

- Explain different statistical methods
- Parametric and non-parametric hypothesis tests
- Design and plan for collecting experimental data
- **Concepts of reliability engineering and the relationship between TTF, R, H**
- How to improve its processes, products and services
- Apply TQM, 6S, and benchmarking



Overview

Identifying best distributions
Fitting distributions to life data
Censored Data fitting
Service and Sales Data fitting

Sources of Failure Data

Suppose we want to quantify the life of a system using time-to-failure data.

Where are the failures from?

Laboratory life testing

- Controlled conditions
 - Loads, Speeds, Environment (Temp, RH)
 - Usually accelerated. 10 years worth of cycles in one month.

Field data

- Uncontrolled conditions
 - Often many unknowns. Subject to external events
 - Takes a long time. 10 years worth of data collecting. All of previous designs.
 - Almost always incomplete. There are field service records of those that failed. Many still in service.

Example: Hydrogen Storage Systems

A typical life test standard. Test conditions to demonstrate system reliability.



SURFACE VEHICLE STANDARD

J2579™ **JUN2018**

Issued 2008-01
Revised 2018-06
Superseding J2579 MAR2013

(R) Standard for Fuel Systems in Fuel Cell and Other Hydrogen Vehicles

RATIONALE

Gaps in requirements for heavy duty hydrogen vehicles have been addressed. See 4.4.2 for hydrogen storage system mounting and installation requirements and 5.2.2.4.1 for guidance in conducting the localized fire test.

The table for pre-approved materials for high pressure hydrogen compatibility in Appendix B.2 was re-formatted to improve understanding. Materials on the table were reviewed and adjusted.

Editorial and format errors were corrected in Appendix B.3 including the re-numbering of sections. Procedures were updated and streamlined throughout Appendix B.3 to improve the quality of test results and expedite verification of materials. Appendix B.3.3 was deleted as a normative requirement as recent test data indicate that Appendix B3.2 is adequate for assessment of fatigue crack growth. References were also added in Appendix B.3 to a newly published standard, CSA CHMC 1, for the verification of the compatibility of materials with compressed hydrogen.

The localized fire test in Appendix C.12 was updated to harmonized acceptance criteria with the approved GTR for hydrogen and fuel cell vehicles. Additional requirements for the test configuration have also been added to address variability in test results that have noted in recent testing.

Requirements for burst ratios of the containers in the Compressed Hydrogen Storage System (CHSS) were modified in Appendix H. Both the minimum allowable prescribed burst ratio and the test method to empirically determine the minimum acceptable value were updated.

FOREWORD

Vehicles manufactured with liquid hydrocarbon as fuels have a long history of implementing appropriate safety measures as specified in SAE Recommended Practices and Standards. With the onset of hydrogen-fueled vehicles, new safety design guidance and methods to verify safe performance will need to be provided to vehicle developers. This SAE Standard addresses hydrogen and hydrogen handling systems on-board vehicles for the purpose of storing, containing, and delivering hydrogen fuel (as defined in SAE J2719) to power generating systems such as fuel cells and internal combustion engines.

This standard provides initial information to be used in the design and construction of hydrogen storage and handling systems to minimize hazards in their operation and maintenance. This document also provides performance-based test criteria for design qualification (performance verification) of hydrogen storage and handling systems in on-road vehicles.

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SAE values your input. To provide feedback on this Technical Report, please visit http://standards.sae.org/J2579_201806

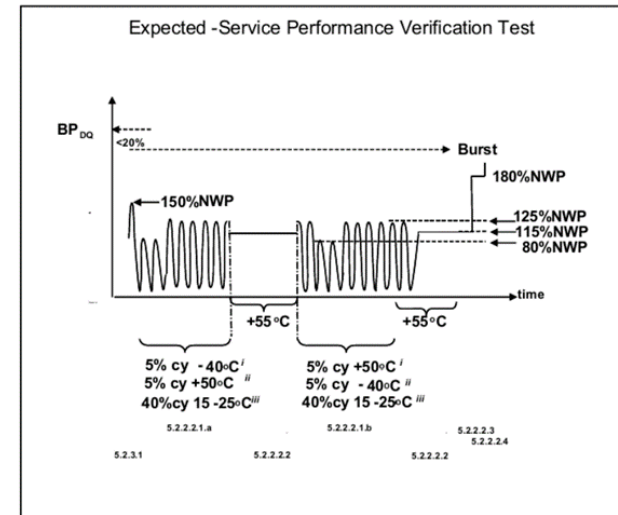


Figure 4 - Compressed hydrogen storage expected-service performance verification

http://standards.sae.org/J2579_201806

Example: Automotive Systems

A typical field life dataset. Some parts failed at a time, some survived to a time.

materials **MDPI**

Article
Reliability Analysis for Unrepairable Automotive Components

Dariusz Ulbrich ¹, Jarosław Selech ^{1,*}, Jakub Kowalczyk ¹, Jakub Józwiak ², Karol Durczak ^{3,4}, Leszek Gil ⁴, Daniel Pieniak ⁵, Marta Paczkowska ¹ and Krzysztof Przytycki ^{6,*}

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⁵ The Main School of Fire Service, Faculty of Safety Engineering and Civil Protection, Słowackiego 52/54, 01-629 Warsaw, Poland; daniel.pieniak@szsp.gov.pl
⁶ Department of Automation, Lublin University of Technology, Nadbystrzycka 6, 20-618 Lublin, Poland
 * Correspondence: jaroslaw.selech@pwr.edu.pl (J.S.); k.przytycki@pwr.edu.pl (K.P.)

Abstract: The analysis of the reliability parameters of a technical object and the determination of the change in the reliability of the object over time, requires the knowledge of the functional characteristics and reliability parameters of the elements included in a system. On the basis of the failure data of the selected element of the object, in this case the vehicle, it is possible to determine the average working time to failure of the element and the appropriate form of distribution that characterizes the reliability and durability parameters of the tested element. The main purpose of the research presented in the article was to develop a method of assessing the reliability of an electronic component of a vehicle—a boot lid contactor. This paper also presents three possible methods of repairing the boot lid contactor (sealing the housing with adhesive with better way, replacing the element with a new one or the most time-consuming solution, changing the shape of the boot lid). The authors also decided to determine the reliability and cost parameters that will allow preventive replacement of this element. The tests were carried out on a fleet of 61 vehicles of the same model, but with different body structures. Contactor failures were recorded in 41 cases, of which 29 were in the hatchback construction and 12 in the estate type. The analysis of the distribution selection for the tested part of the passenger car the boot lid contactor was performed using the Likelihood Value (LKV) test to determine the rank of distributions. Also the maximum likelihood (MLE) method was used to estimate the distribution parameters. The three-parameter Weibull distribution was the best fitted distribution in both cases. It was clearly defined that one model of car with two different types of body have vastly different reliability characteristic. Based on the reliability characteristic and parameters, the appropriate preventive actions can be taken, minimizing the risk of damage, thus avoiding financial losses and guaranteeing an appropriate level of vehicle safety.

Keywords: reliability; failure analysis; automotive

1. Introduction

In the reliability analysis of technical objects, various probability distributions can be used for modeling damage data [1–3]. Nevertheless, the most commonly used probability distributions are the normal, exponential, Weibull or gamma ones [4–6]. In the research results presented in this paper, for the reliability analysis of the boot lid contactor, apart from the above distributions, other, less frequently used distributions were used, the quality of which, for the analyzed case, is better than the usually used distributions. Less frequently used distributions include the log-normal distribution, the generalized gamma distribution,

check for updates

Citation: Ulbrich, D.; Selech, J.; Kowalczyk, J.; Józwiak, J.; Durczak, K.; Gil, L.; Pieniak, D.; Paczkowska, M.; Przytycki, K. Reliability Analysis for Unrepairable Automotive Components. *Materials* **2023**, *16*, 7014. <https://doi.org/10.3390/mat16227014>

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Materials **2023**, *16*, 7014. <https://doi.org/10.3390/mat16227014> <https://www.mdpi.com/journal/materials>

Table 1. Failure/suspension date of contactor (F/S—failure, suspension, car body type: H—hatchback, E—Estate).

F/S	Mileage [km]	Car Body Type	F/S	Mileage [km]	Car Body Type	F/S	Mileage [km]	Car Body Type
F	52,621	H	F	126,639	E	F	272,945	H
F	57,801	H	F	128,676	H	S	21,147	E
F	58,001	H	F	133,390	H	S	21,169	E
F	61,628	H	F	143,159	E	S	21,772	E
F	68,100	H	F	144,041	H	S	23,071	E
F	79,051	H	F	151,230	H	S	27,165	H
F	85,974	H	F	153,345	E	S	36,019	E
F	87,451	E	F	158,420	E	S	40,208	E
F	88,050	H	F	161,200	H	S	44,374	H
F	92,003	E	F	163,141	H	S	45,895	E
F	95,636	H	F	163,952	E	S	49,990	E
F	102,178	H	F	169,230	H	S	63,422	E
F	105,600	H	F	176,965	E	S	63,519	H
F	106,639	H	F	193,500	H	S	86,527	H
F	110,558	H	F	199,263	E	S	99,784	E
F	113,359	H	F	204,898	E	S	120,111	E
F	115,000	H	F	206,460	E	S	128,523	E
F	115,210	H	F	214,521	H	S	142,302	E
F	116,762	E	F	231,403	H	S	151,209	E
F	125,489	H	F	253,241	H	S	164,287	E
						S	171,356	H

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8619900/pdf/materials-14-07014.pdf>

Distribution Fitting

Suppose we seek to fit a distribution to a sample of life data.

What distribution? Weibull?

$$f(t) = \frac{\beta}{\eta} \left(\frac{t}{\eta} \right)^{\beta-1} e^{-\left(\frac{t}{\eta} \right)^{\beta}}$$

What is the best fit β and η to a set of data?

What are their confidence intervals?

Life
4.23
3.98
4.36
3.86
4.12
4.23
4.36
4.44
3.98
4.06
3.79
4.35
4.20
4.57
4.82
4.44
4.94
4.68
4.57
4.44
4.36
4.82
4.74
5.01
4.45



Component Life Distributions

Weibull

- A very useful distribution that can be used to model many types of failure data. Can model early life failures, random normal-life failures and wear out end-of-life.

Lognormal

- A good model for failures driven by diffusion equation processes such as corrosion and migration. Electronic device components.

Exponential

- A good model for random external event normal-life failures. Not a wear-out end-of-life distribution. Also fits the life of systems comprised of large numbers of things (why?).



Component Life Distributions

Reliability analysis tools provide a library of alternative distribution functions that can be fit to life data. Which one should you use?

Often we try them all and pick the one that fits best.

Try not to do this. Failure mode physics determines the life distribution.

The hazard rate is determined by the physics of failure process.

Is it manufacturing related early-life, random normal-life, or wear-out end-of-life?

Some distributions are better representative of each of these.

Maximum Likelihood Regression

Recall we use maximum likelihood regression to fit a distribution to life data.

For a probability density function $f(\theta, t)$, the likelihood function for a sample of life data t_i is give by

$$L(\theta) = \prod_{i=1}^n f(\theta, t_i)$$

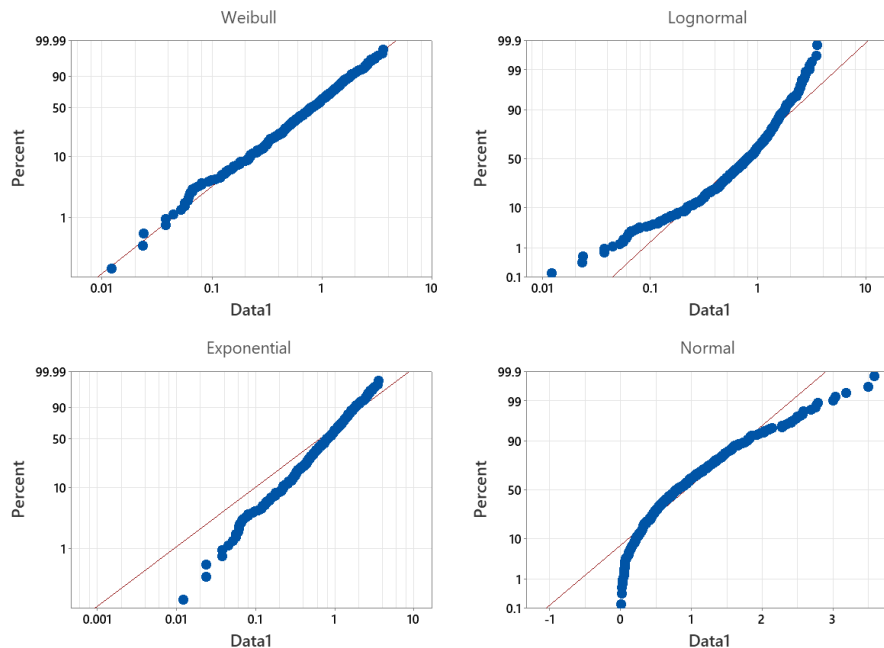
Finding the best parameter estimates for the distribution function is in general an optimization problem, generally taking the $\ln()$ for smoother behaviour:

$$\text{Find } \theta^* = \underset{\theta}{\operatorname{argmax}} \ln(L(\theta))$$

Distribution Identification Fitting

Fit a selection of alternative candidate distribution functions.

Look at the results graphically and judge which fits better or if the data is acceptable.



Distribution Identification Fitting – Minitab

Explore alternative distributions using

Stat > Reliability/Survival/Distribution Analysis (Right Censoring) > Distribution ID Plot

Distribution ID Plot-Right Censoring

Variables:
Data1

Frequency columns (optional):

☐ By variable:

☐ Use all distributions

☒ Specify

☒ Distribution 1: Weibull

☒ Distribution 2: Lognormal

☒ Distribution 3: Exponential

☒ Distribution 4: Normal

Select

Help

Censor...

Options...

OK

Cancel

Data1
0.221153
0.936777
0.504052
0.744562
1.117697
1.833689
1.83041
1.080317
1.497522
1.462568
0.447256
:

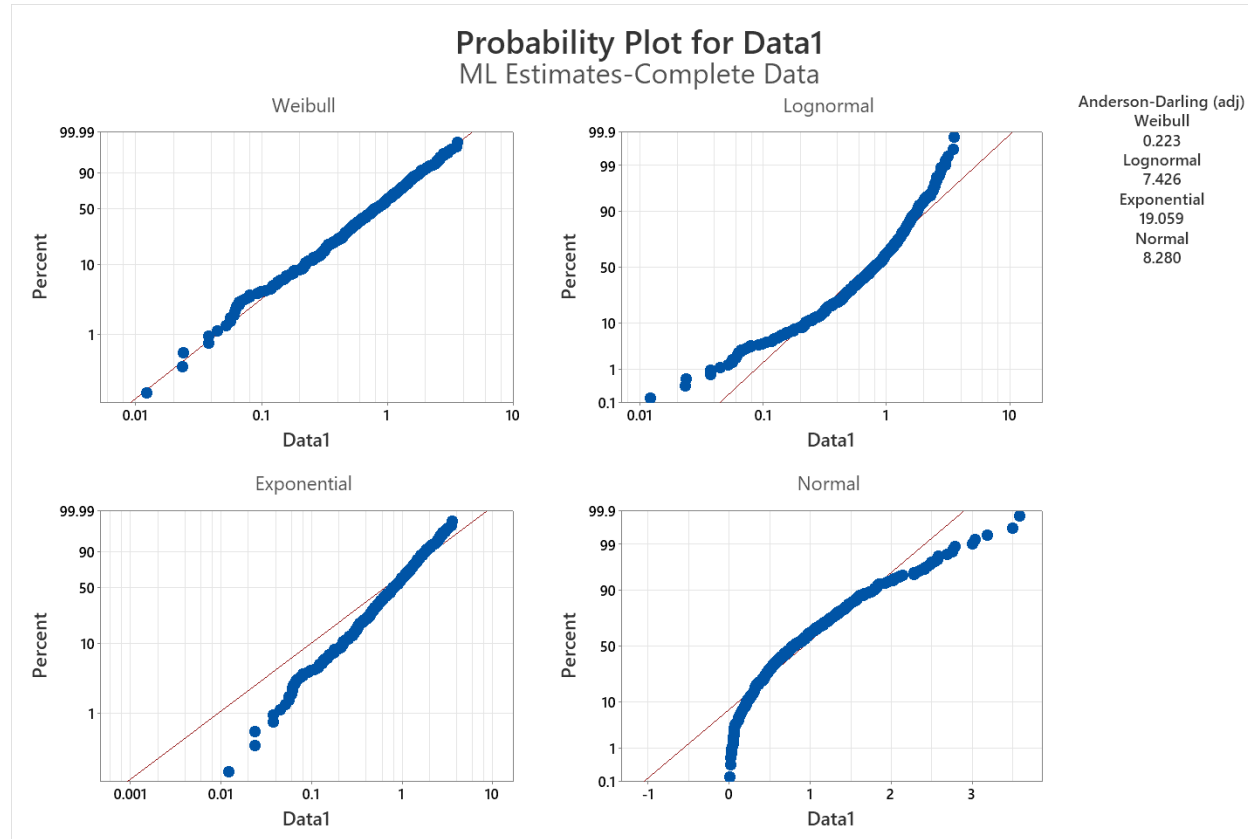
Distribution Identification Fitting – Python

Explore alternative distributions using

```
raw_data = pd.read_excel("Data01.xlsx")
Fit_Everything(failures=raw_data['Data1'].values)
```

Distribution	Alpha	Beta	Gamma	Mu	Sigma	Lambda	Log-likelihood	AICc	BIC	AD
Weibull_2P	1.02019	1.46299					-411.624	827.272	835.677	0.223497
Weibull_3P	1.02019	1.46299	0				-411.624	829.296	841.892	0.223497
Gamma_2P	0.506582	1.82493					-415.494	835.012	843.417	0.895538
Gamma_3P	0.506582	1.82493	0				-415.494	837.036	849.631	0.895538
Loglogistic_2P	0.747003	2.09339					-445.691	895.406	903.812	3.92737
Loglogistic_3P	0.747003	2.09339	0				-445.691	897.431	910.026	3.92737
Exponential_2P			0.0121947			1.09615	-454.097	912.218	920.623	17.4088
Exponential_1P						1.08169	-460.736	923.48	927.687	19.059
Lognormal_2P			-0.376866	0.885766			-460.385	924.794	933.199	7.4262
Lognormal_3P			0	-0.376866	0.885766		-460.385	926.818	939.414	7.4262
Normal_2P			0.924476	0.639451			-485.897	975.818	984.223	8.27988
Gumbel_2P			1.27268	0.796233			-604.727	1213.48	1221.88	26.7081

Which Distribution Fits Well?



Goodness of Fit – Minitab

The Goodness-of-Fit indicator shown by Minitab is the Anderson-Darling statistic.
Smaller AD indicates a better fit.

Goodness-of-Fit	
Distribution	Anderson-Darling (adj)
Weibull	0.223
Lognormal	7.426
Exponential	19.059
Normal	8.280

Goodness of Fit – Python

The Goodness-of-Fit indicator shown by Python include the Log-Likelihood optimization function, the AICc, BIC and the Anderson-Darling statistics.

Smaller indicates a better fit.

Distribution	Alpha	Beta	Gamma	Mu	Sigma	Lambda	Log-likelihood	AICc	BIC	AD
Weibull_2P	1.02019	1.46299					-411.624	827.272	835.677	0.223497
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Overview

Identifying best distributions

Fitting distributions to life data

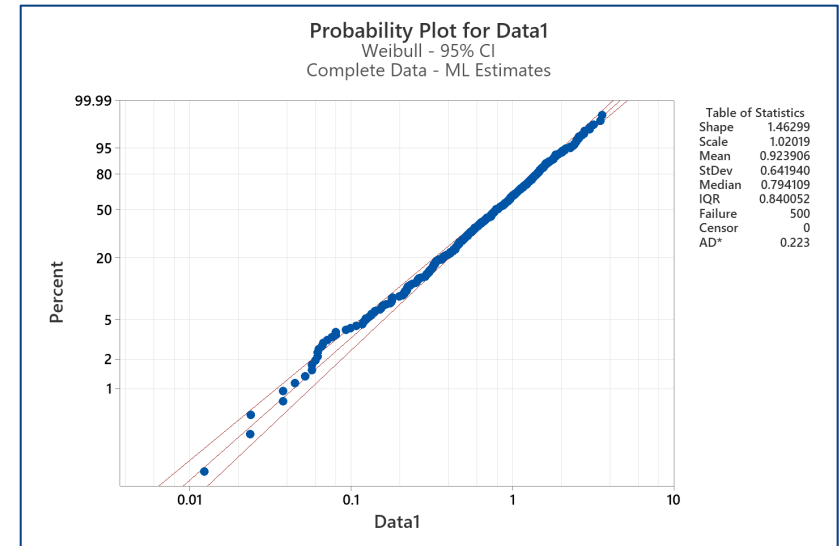
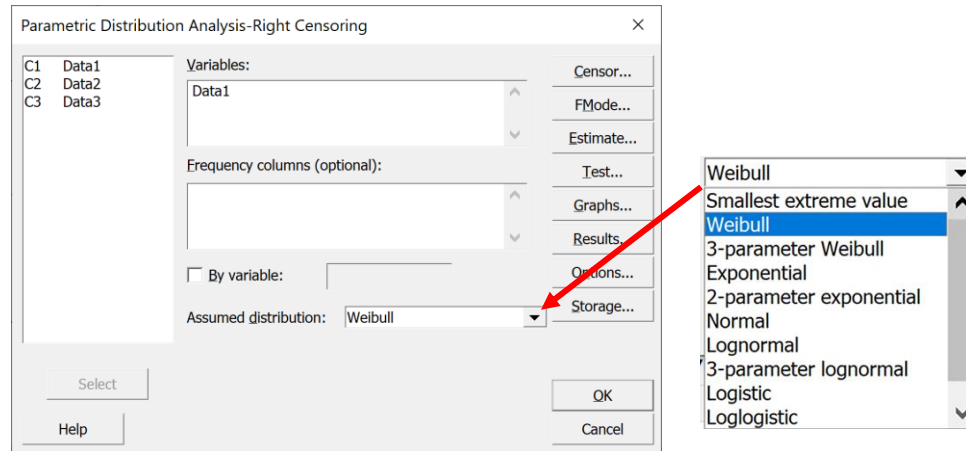
Censored Data fitting

Service and Sales Data fitting

Distribution Fitting – Minitab

With a selected distribution function, fitting in Minitab is in

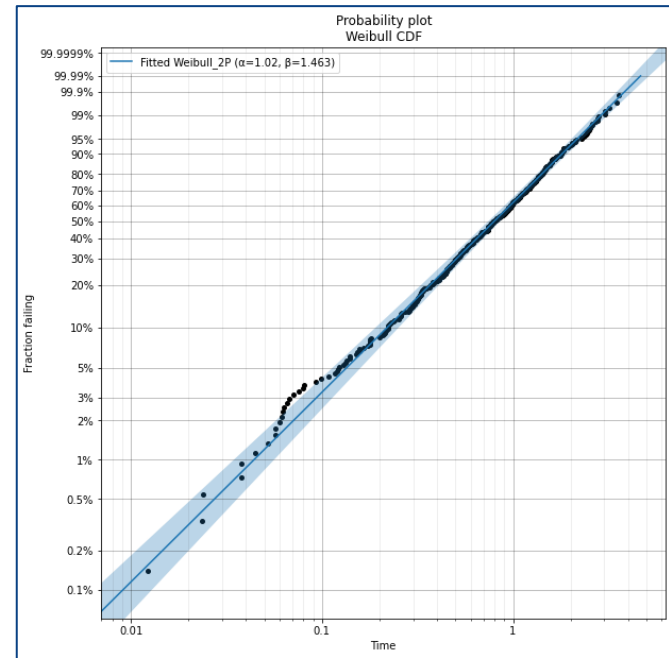
Stat > Reliability/Survival/Distribution Analysis > Parametric Distribution Analysis



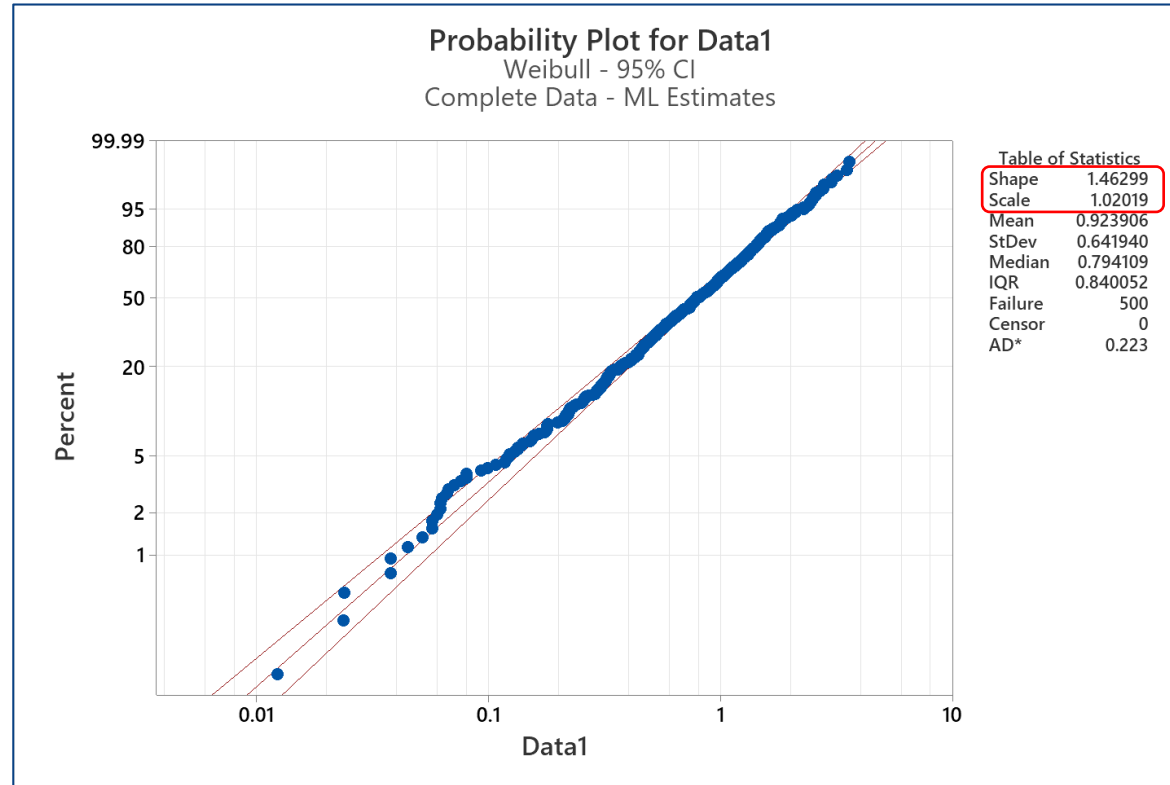
Distribution Fitting – Python

Available in the reliability code library

```
raw_data = pd.read_excel("Data01.xlsx")  
wb = Fit_Weibull_2P(failures=raw_data['Data1'].values)  
plt.show()
```

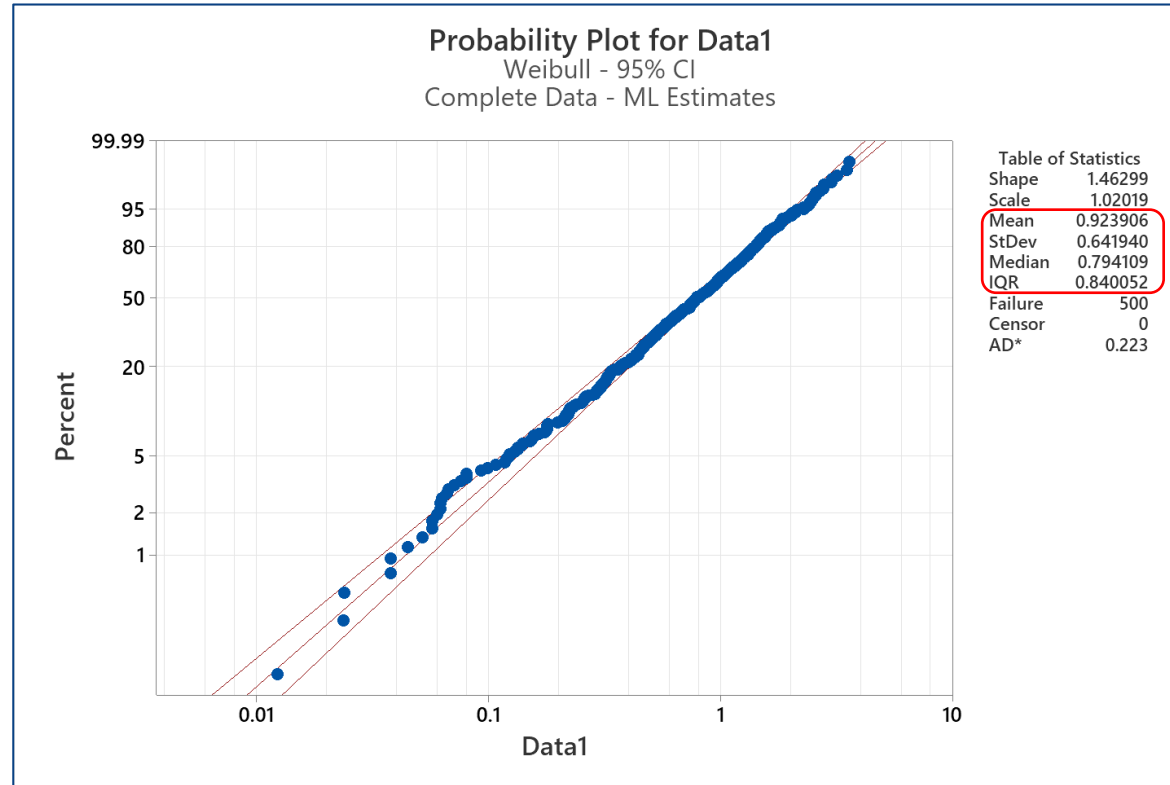


Distribution Fit



Best fit distribution
parameters (β, η)

Distribution Fit



Sample statistics

MLE Fit Table

Similar to a regression table, the confidence intervals on the parameters

Parameter Estimates

Parameter	Estimate	Standard Error	95.0% Normal CI	
			Lower	Upper
Shape	1.46299	0.0511349	1.36612	1.56672
Scale	1.02019	0.0328184	0.957853	1.08658

Also further statistics

Characteristics of Distribution

	Estimate	Standard Error	95.0% Normal CI	
			Lower	Upper
Mean (MTTF)	0.923906	0.0286720	0.869385	0.981847
Standard Deviation	0.641940	0.0261664	0.592650	0.695329
Median	0.794109	0.0284884	0.740190	0.851955
First Quartile (Q1)	0.435341	0.0218425	0.394568	0.480327
Third Quartile (Q3)	1.27539	0.0390893	1.20104	1.35435
Interquartile Range(IQR)	0.840052	0.0294871	0.784202	0.899880

MLE Fit Table

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Also further statistics

What type of failure mode is this?

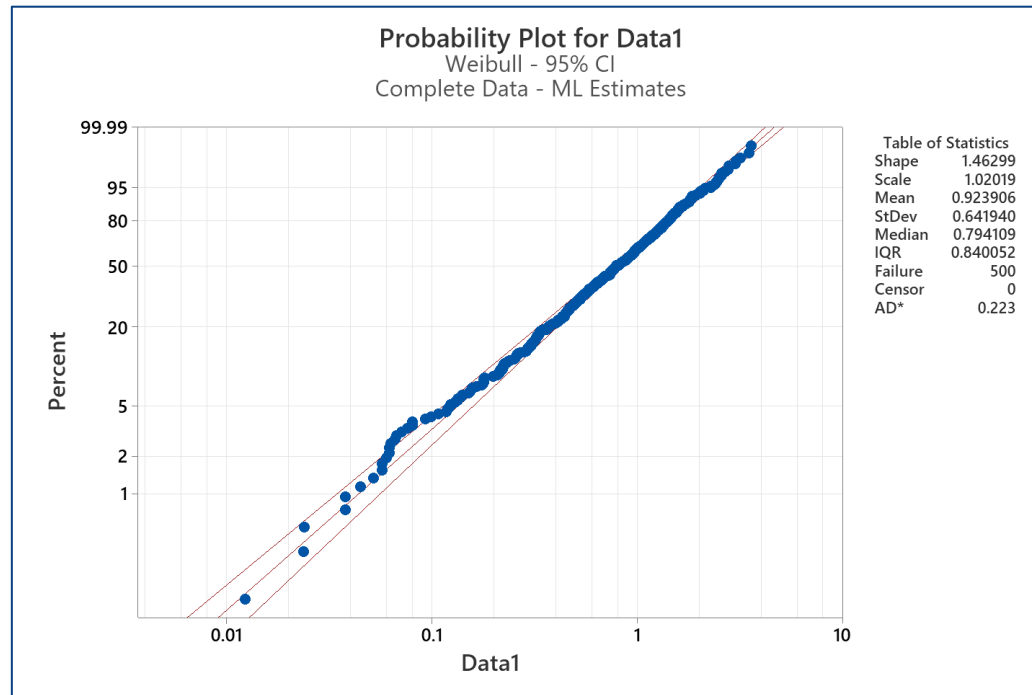
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Life Confidence Interval

Suppose your company is willing to accept a 1% failure rate in a warranty period.

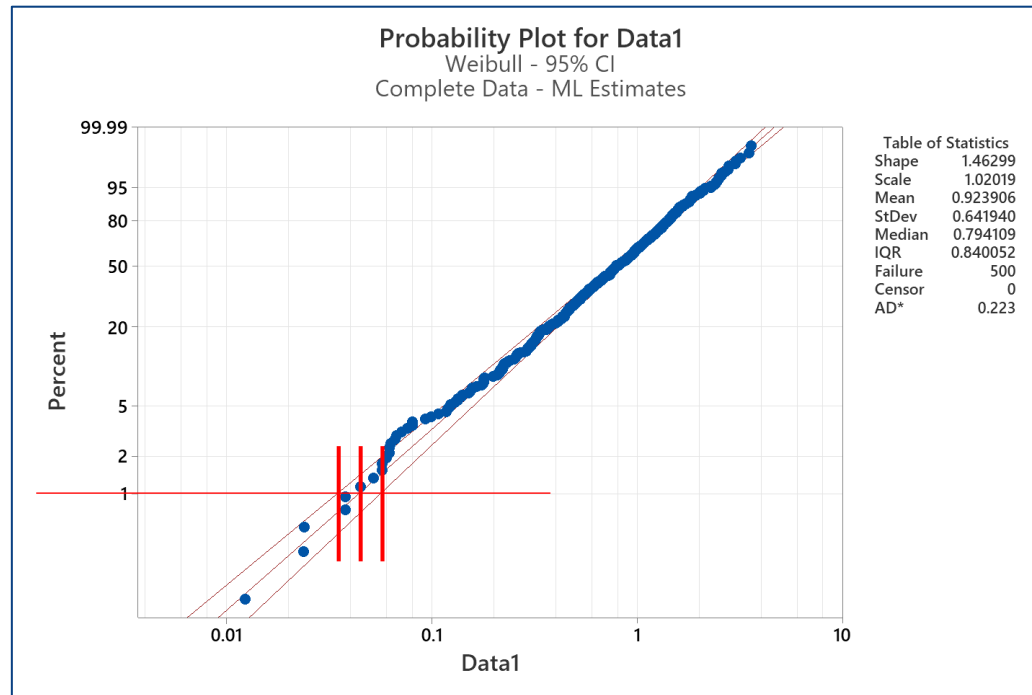
What is the confidence interval for when the first 1% have failed?



Life Confidence Interval

Suppose your company is willing to accept a 1% failure rate in a warranty period.

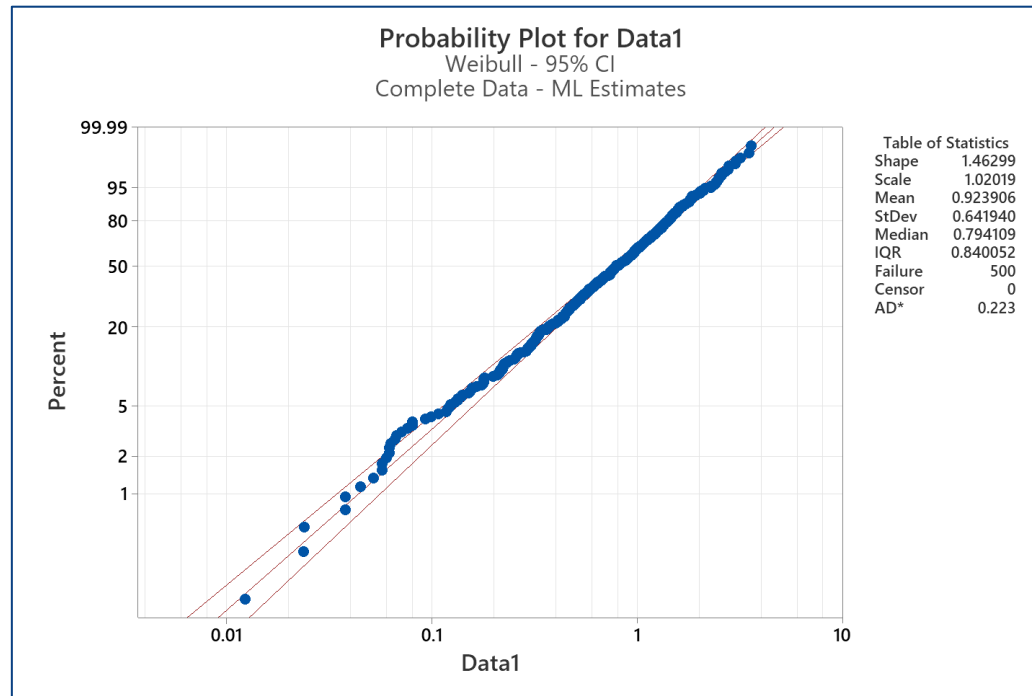
What is the confidence interval for when the first 1% have failed?



Reliability Confidence Interval

Suppose your company has defined the warranty period as 0.1 years.

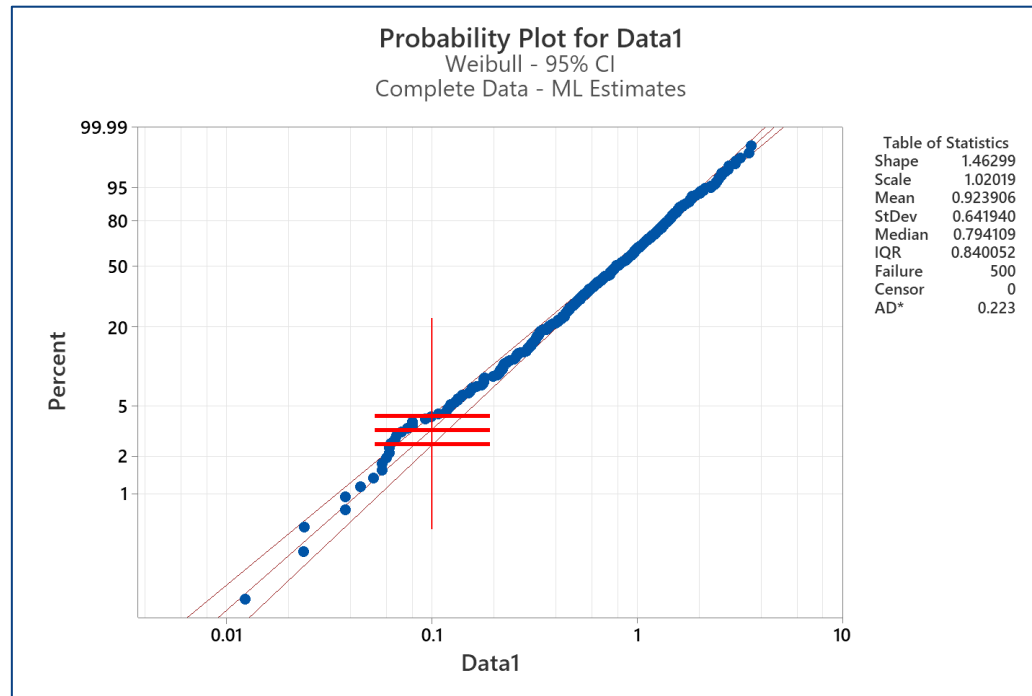
What is the confidence interval for the reliability?



Reliability Confidence Interval

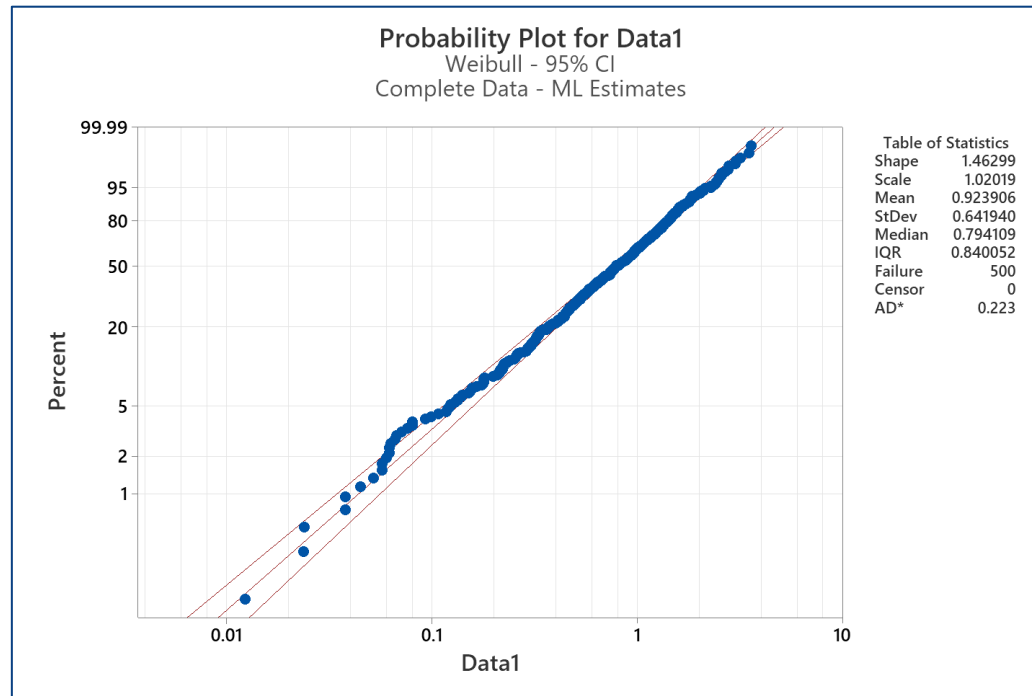
Suppose your company has defined the warranty period as 0.1 years.

What is the confidence interval for the reliability?



Percentile Confidence Intervals

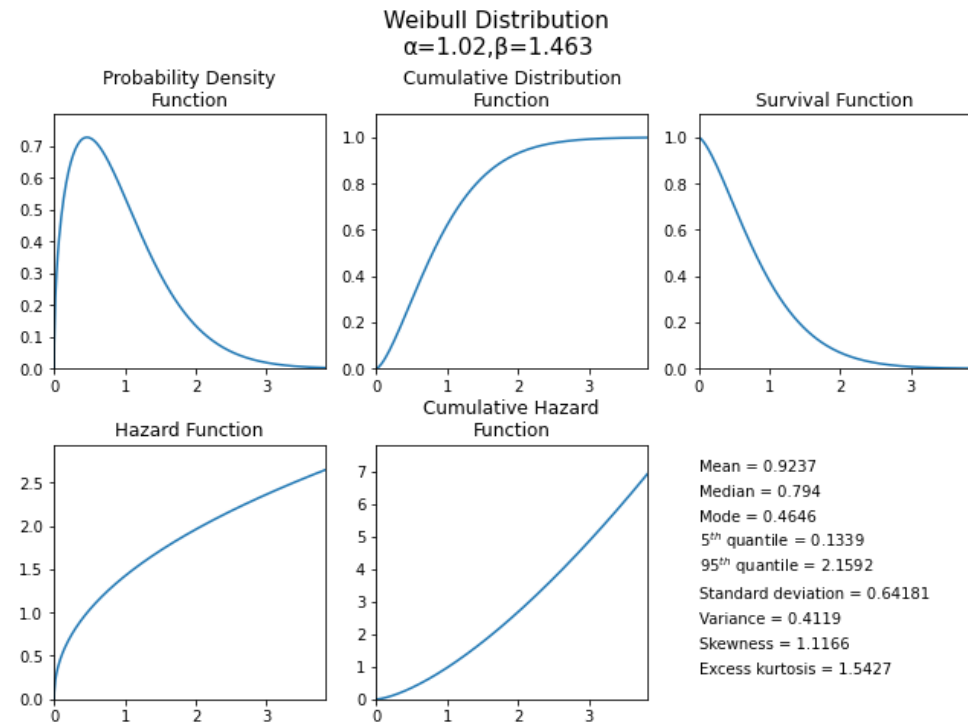
The 95% confidence intervals are for the percentiles fit. The line.
They are not prediction intervals for all future data.



Reliability Distribution Graphs

With a selected distribution function, the four reliability plots are useful to plot.

- Probability density function
- Probability Plot (cumulative density)
- Survival function (reliability)
- Hazard function



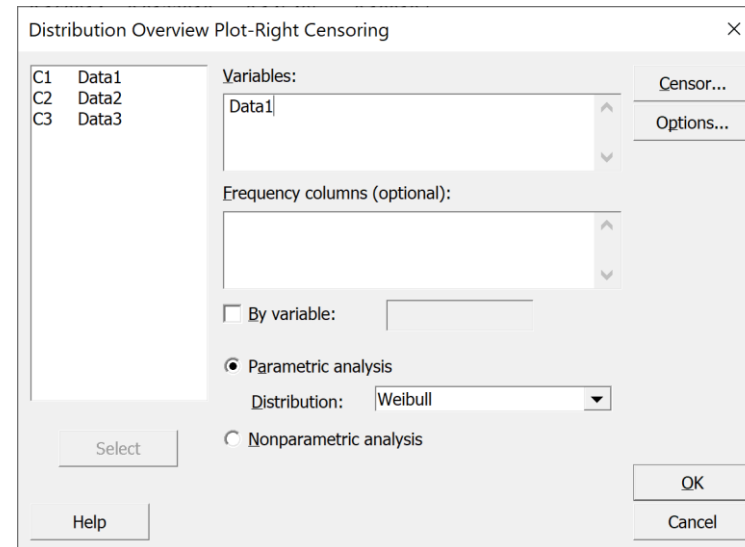
Reliability Distribution Graphs

With a selected distribution function, the four reliability plots in Minitab are

Stat > Reliability/Survival/Distribution Analysis > Distribution Overview Plot

Shows four important plots

- Probability density function
- Probability Plot (cumulative density)
- Survival function (reliability)
- Hazard function



Distribution Plots – Minitab

Density

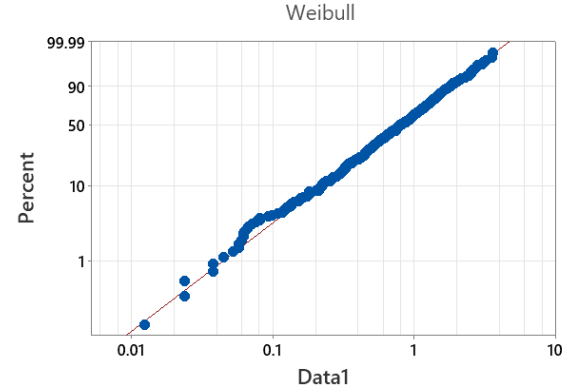
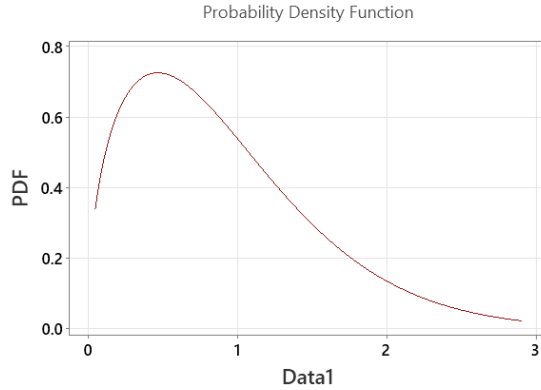
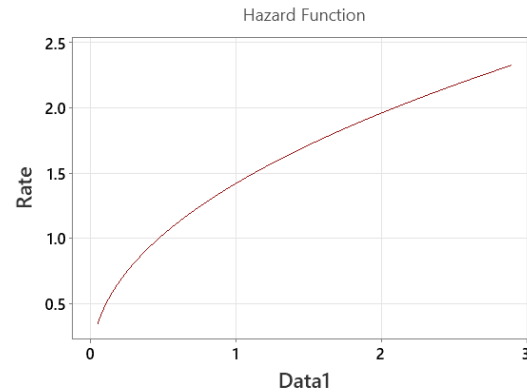
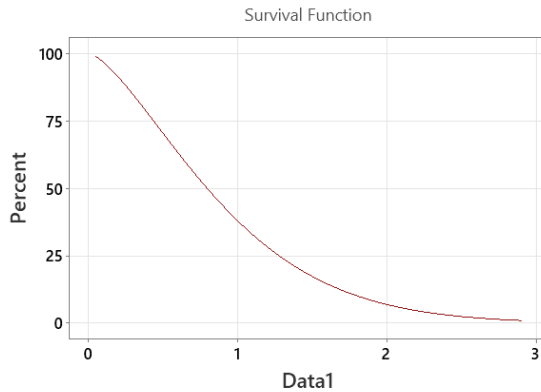


Table of Statistics	
Shape	1.46299
Scale	1.02019
Mean	0.923906
StDev	0.641940
Median	0.794109
IQR	0.840052
Failure	500
Censor	0
AD*	0.223

Survival

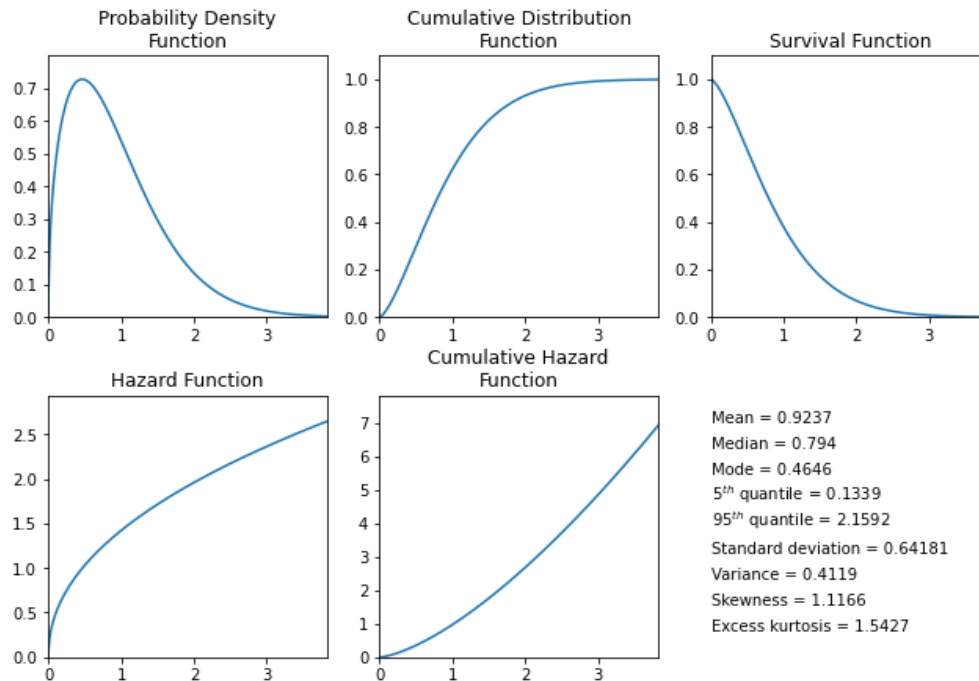


Hazard

Distribution Plots – Python

```
from reliability.Distributions import Weibull_Distribution  
  
dist = Weibull_Distribution(alpha=1.02,beta=1.463)  
dist.plot()
```

Weibull Distribution
 $\alpha=1.02, \beta=1.463$



Exercise

Use the data in “L11 Reliability Data.xlsx”

Columns “Data1”, “Data2” and “Data3”

What distributions fit the data?

- Write down the parametric equation.
- What is the MTBF?
- What is the time until the first 1% have failed?

Distribution ID Plot-Right Censoring

Variables:
Data1

Frequency columns (optional):

☐ By variable:

☐ Use all distributions

☒ Specify

☒ Distribution 1: Weibull

☒ Distribution 2: Lognormal

☒ Distribution 3: Exponential

☒ Distribution 4: Normal

Select

Help

OK

Cancel

Goodness-of-Fit

Distribution Anderson-Darling (adj)

Weibull	0.223
Lognormal	7.426
Exponential	19.059
Loglogistic	3.927
3-Parameter Weibull	0.264
3-Parameter Lognormal	0.789
2-Parameter Exponential	17.638
3-Parameter Loglogistic	1.538
Smallest Extreme Value	26.708
Normal	8.280
Logistic	5.772

Parametric Distribution Analysis-Right Censoring

Variables:
Data1

Frequency columns (optional):

☐ By variable:

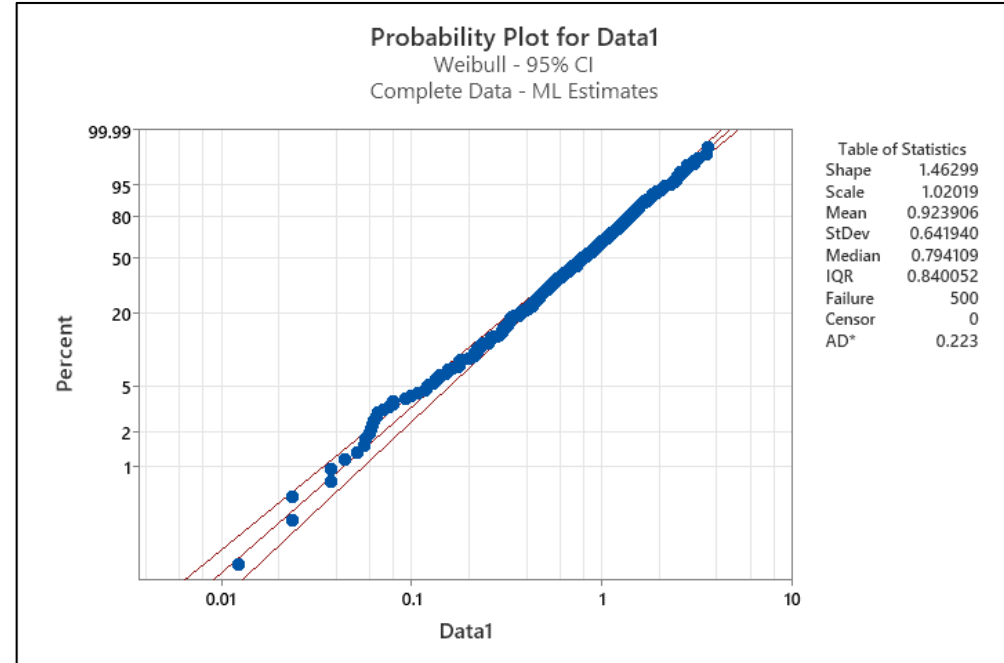
Assumed distribution: Weibull

Select

Help

Censor...
FMode...
Estimate...
Test...
Graphs...
Results...
Options...
Storage...

OK
Cancel



Parameter Estimates

Parameter	Estimate	Standard Error	95.0% Normal CI	
			Lower	Upper
Shape	1.46299	0.0511349	1.36612	1.5667
Scale	1.02019	0.0328184	0.957853	1.0865

Characteristics of Distribution

	Estimate	Standard Error	95.0% Normal CI	
			Lower	Upper
Mean(MTTF)	0.923906	0.0286720	0.869385	0.98184
Standard Dev	0.641940	0.0261664	0.592650	0.69532
Median	0.794109	0.0284884	0.740190	0.85195
1st Quartile(Q1)	0.435341	0.0218425	0.394568	0.48032
3rd Quartile(Q3)	1.27539	0.0390893	1.20104	1.35435
InterQ Range(IQR)	0.840052	0.0294871	0.784202	0.89988

Table of Percentiles

%	Percentile	Standard Error	95.0% Normal CI	
			Lower	Upper
1	0.0439646	0.0054410	0.0344954	0.0560333
2	0.0708551	0.0076291	0.0573746	0.0875028
3	0.0938104	0.0092222	0.0773700	0.113744
4	0.114599	0.0105068	0.0957501	0.137158
5	0.133956	0.0115951	0.113054	0.158724
6	0.152280	0.0125451	0.129574	0.178964
7	0.169814	0.0133910	0.145495	0.198197
8	0.186722	0.0141554	0.160941	0.216633
9	0.203123	0.0148537	0.176000	0.234425
10	0.219103	0.0154973	0.190740	0.251683
20	0.365945	0.0201674	0.328478	0.407687
30	0.504248	0.0232997	0.460588	0.552046
40	0.644578	0.0258957	0.595770	0.697384
50	0.794109	0.0284884	0.740190	0.851955
60	0.961014	0.0315976	0.901037	1.02498
70	1.15820	0.0359908	1.08977	1.23094
80	1.41237	0.0432163	1.33016	1.49967
90	1.80413	0.0579731	1.69401	1.92141
91	1.86016	0.0604160	1.74543	1.98242
92	1.92188	0.0631953	1.80193	2.04982
93	1.99076	0.0664023	1.86478	2.12526
94	2.06894	0.0701707	1.93588	2.21114
95	2.15967	0.0747086	2.01810	2.31117
96	2.26837	0.0803652	2.11620	2.43148
97	2.40506	0.0877969	2.23899	2.58344
98	2.59184	0.0984825	2.40583	2.79223
99	2.89756	0.117159	2.67679	3.13653



Overview

Identifying best distributions

Fitting distributions to life data

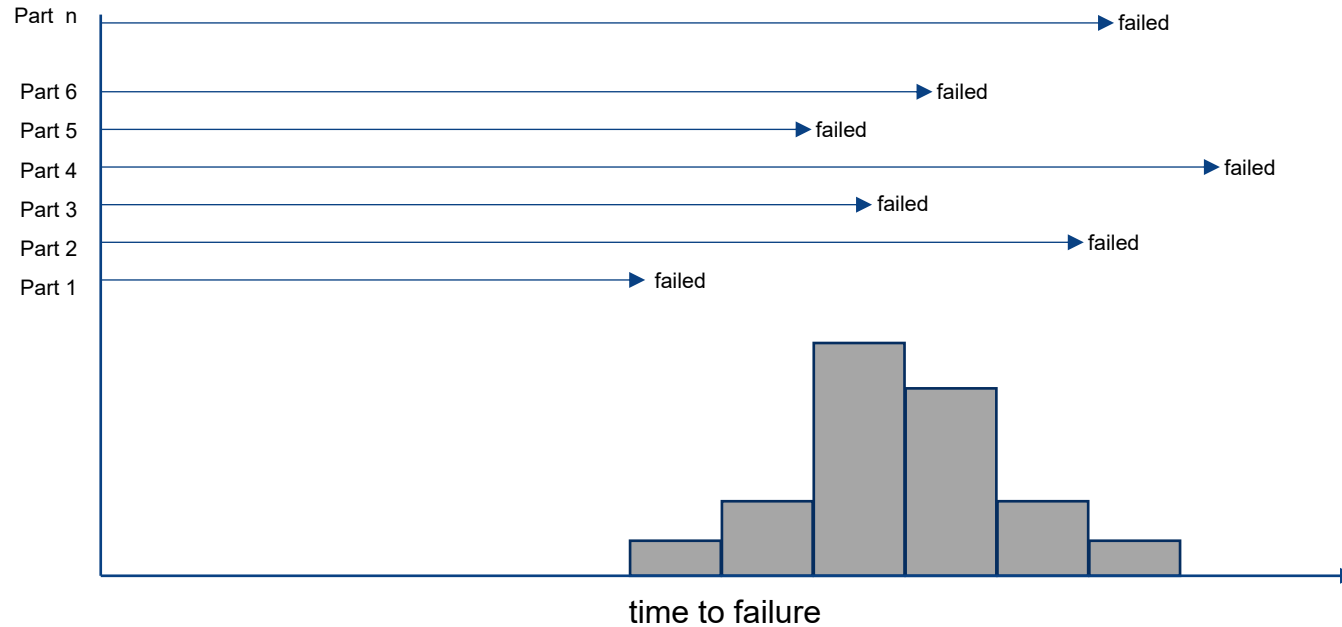
Censored Data fitting

Service and Sales Data fitting

Test to Failure

The most simple life testing data is when every sample trial is run until it fails.

Also called ***uncensored data***.



Uncensored Data

Suppose you have tested 10 components to failure.

You get back following life data.

What is your estimate of the average life?

Life (hrs)
323
389
453
510
655
702
1093
1487
1594
1734
1987

Uncensored Data

Suppose you have tested 10 components to failure.

You get back following life data.

What is your estimate of the average life?

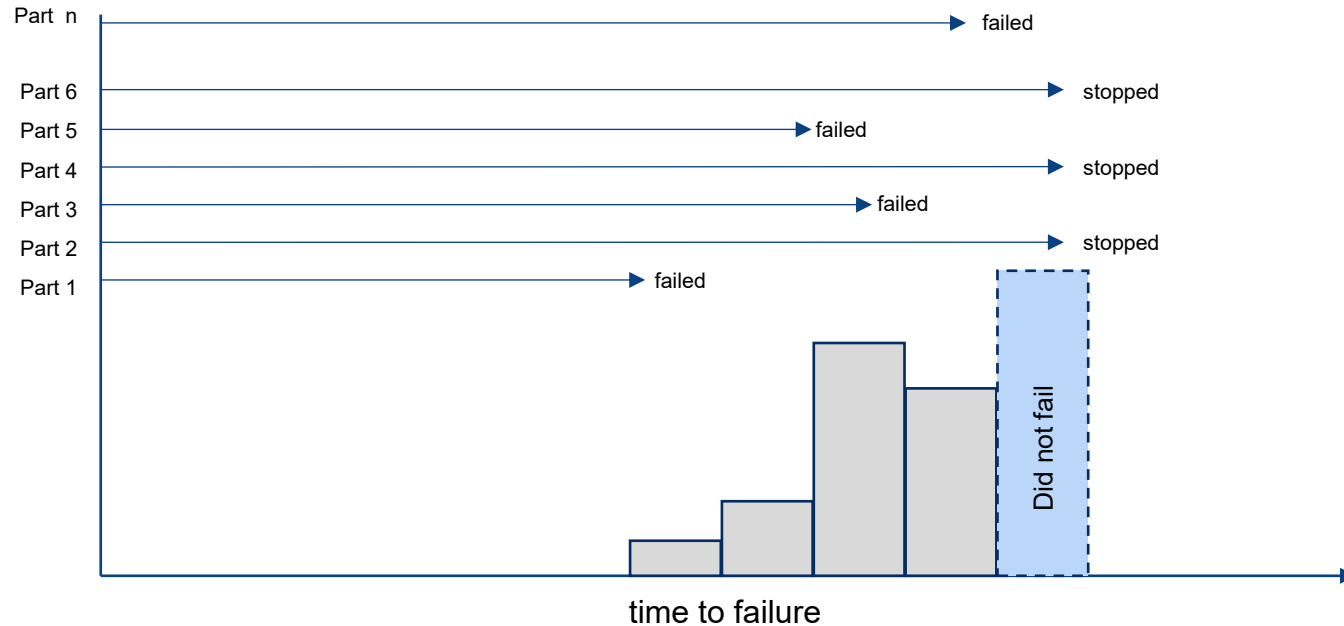
$$\begin{aligned}\text{MTTF} &= \frac{1}{n} \sum L_i \\ &= 993 \text{ hours}\end{aligned}$$

Life (hrs)
323
389
453
510
655
702
1093
1487
1594
1734
1987

Test to a Fixed Duration

Now suppose you stop the test after you run out of time. Test stopped before some failed.

Also called *censored data*.



Data

Suppose you have tested 10 components to 1000 hours.

You get back following life data.

What is your estimate of the average life?

Life (hrs)
323
389
453
510
655
702
Did not fail
Did not fail
Did not fail
Did not fail
Did not fail

Data

Suppose you have tested 10 components to 1000 hours.

You get back following life data.

What is your estimate of the average life?

Mistake #1:

$$MTTF = \frac{323+389+453+510+655+702}{6}$$

Life (hrs)
323
389
453
510
655
702
Did not fail
Did not fail
Did not fail
Did not fail
Did not fail

Data

Suppose you have tested 10 components to 1000 hours.

You get back following life data.

What is your estimate of the average life?

Mistake #1:

$$MTTF = \frac{323+389+453+510+655+702}{6}$$

Mistake #2:

$$MTTF = \frac{323+389+453+510+655+702+5*1000}{11}$$

Both underestimate the true life.

Life (hrs)
323
389
453
510
655
702
Did not fail
Did not fail
Did not fail
Did not fail
Did not fail

Maximum Likelihood Regression, Censored Data

For a distribution with uncensored data t_i , the likelihood function was given by

$$L(\theta) = \prod_{i=1}^n f(\theta, t_i)$$

Consider now a censor indicator δ_i that is 1 if the part failed and 0 if it was censored. To contribute to the Lagrangian, all we know is the censored value survived to time t_i . Therefore we include the survival function value $F(\theta, t_i)$ for the censored data,

$$L(\theta) = \prod_{i=1}^n f(\theta, t_i)^{\delta_i} F(\theta, t_i)^{1-\delta_i}$$

Finding these best parameter estimates for the distribution function again an optimization problem:

$$\text{Find } \theta^* = \underset{\theta}{\operatorname{argmax}} L(\theta)$$

Distribution Identification Fitting – Minitab

Retry alternative distributions now with censored data.

Stat > Reliability/Survival/Distribution Analysis (Right Censoring) > Distribution ID Plot

Use the data in “L10 Reliability.xlsx”

Column “CData1” and “Censor 1”

This is life test data in days where the test was stopped at 1 day.

The censor column is 0 if the value is a failure, and a 1 if the value is censored.

CData1	Censor1
0.012295	0
0.02364	0
0.023806	0
0.037749	0
0.037928	0
0.044812	0
0.051941	0
0.056847	0
0.056887	0
:	:
1.000	1
1.000	1
1.000	1

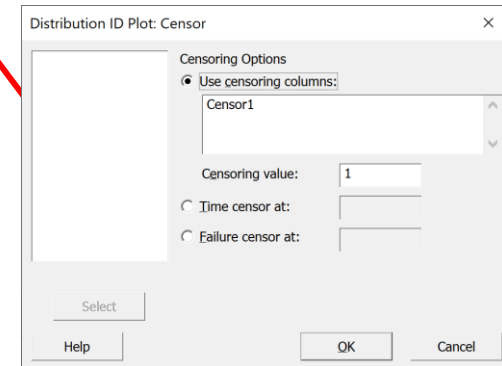
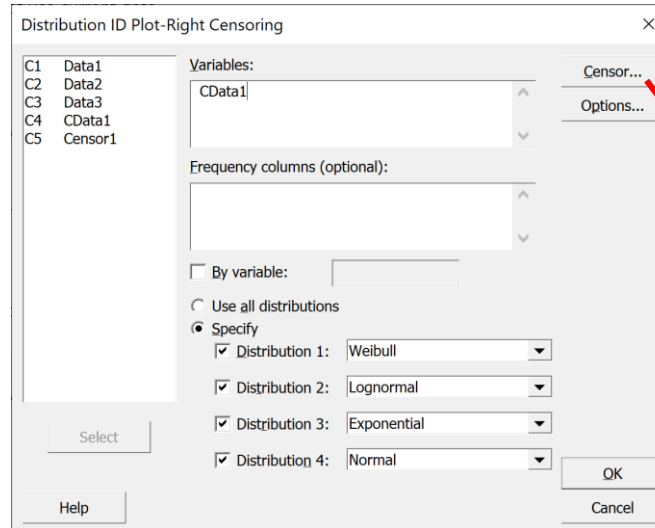
Distribution Identification Fitting – Minitab

Retry alternative distributions now with censored data.

Stat > Reliability/Survival/Distribution Analysis (Right Censoring) > Distribution ID Plot

Select “CData1” for the life data variable.

Select “Censor1” for the censor variable, and define ‘1’ as the censor value.

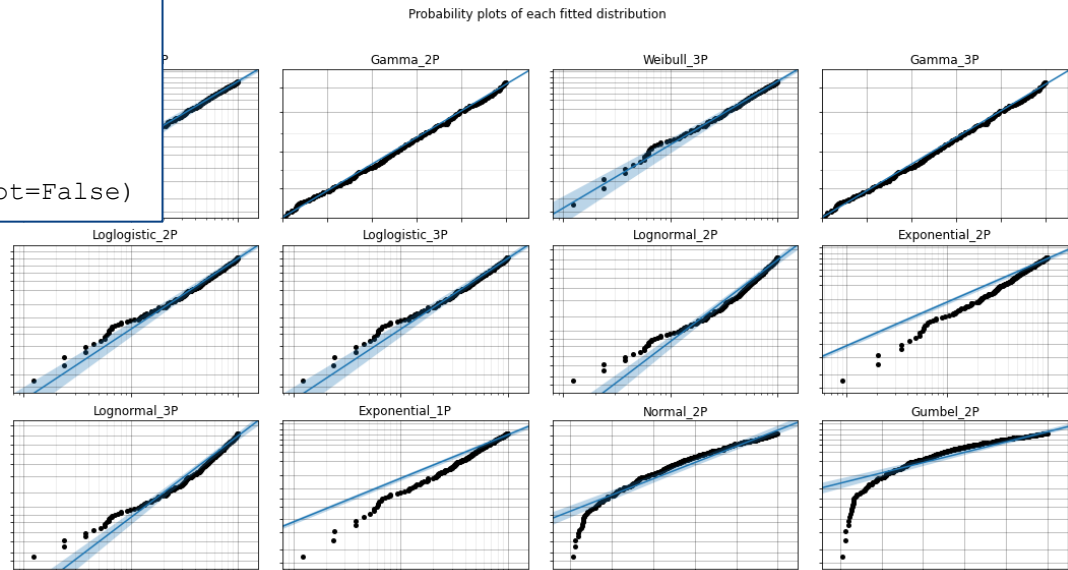


Distribution Identification Fitting – Python

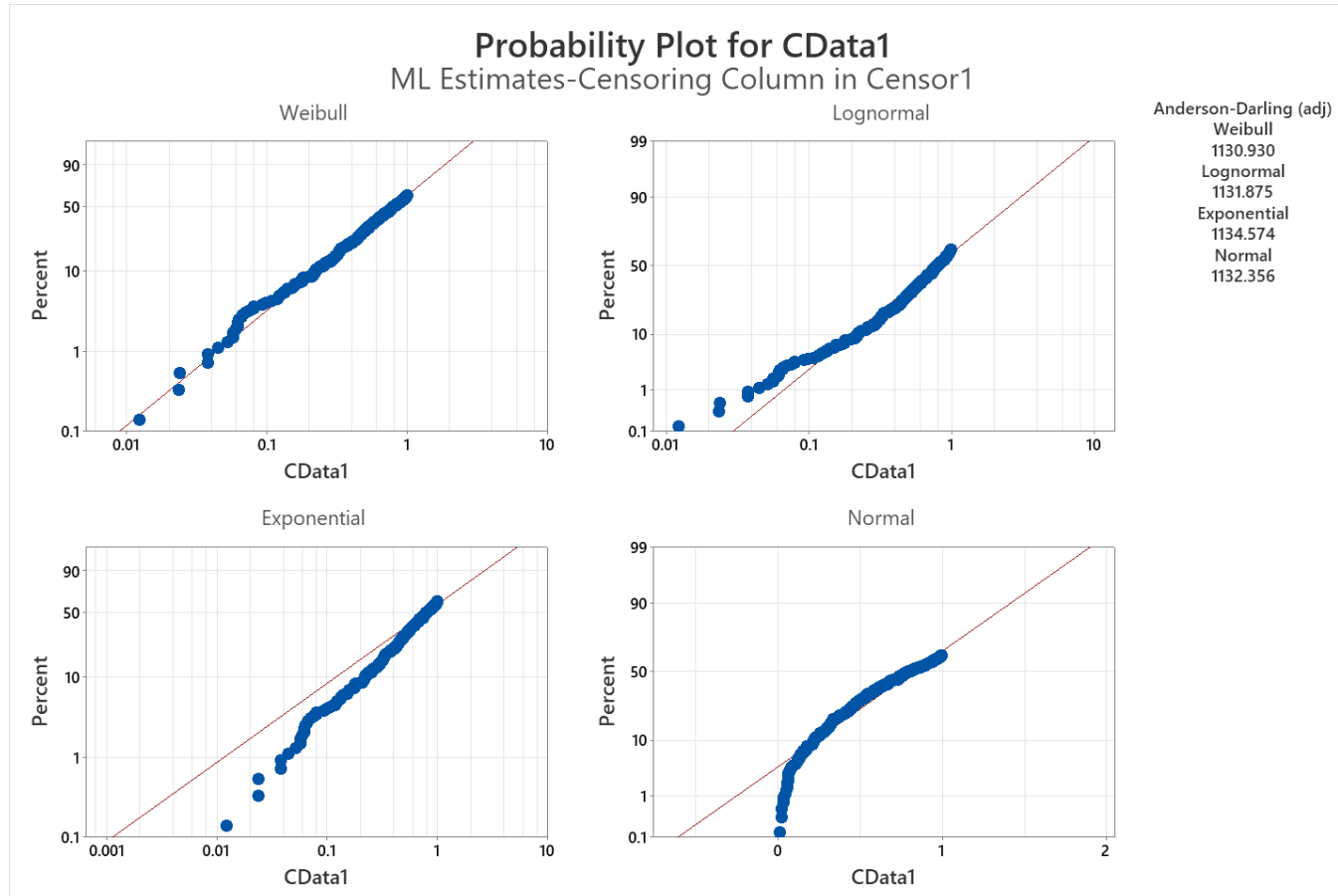
```
from reliability.Fitters import Fit_Everything
from reliability.Convert_data import xlsx_to_XCN
from reliability.Convert_data import XCN_to_FR
```

```
data = xlsx_to_XCN(path=r'CDData01.xlsx')
dataFR = XCN_to_FR(data.X, data.C, data.N)
```

```
Fit_Everything(failures=dataFR.failures,
               right_censored=dataFR.right_censored,
               show_histogram_plot=False,
               show_probability_plot=True,
               show_PP_plot=False,
               show_best_distribution_probability_plot=False)
```



Which Distribution Fits Well?





Goodness of Fit

The Goodness-of-Fit indicator shown by Minitab is the Anderson-Darling statistic.

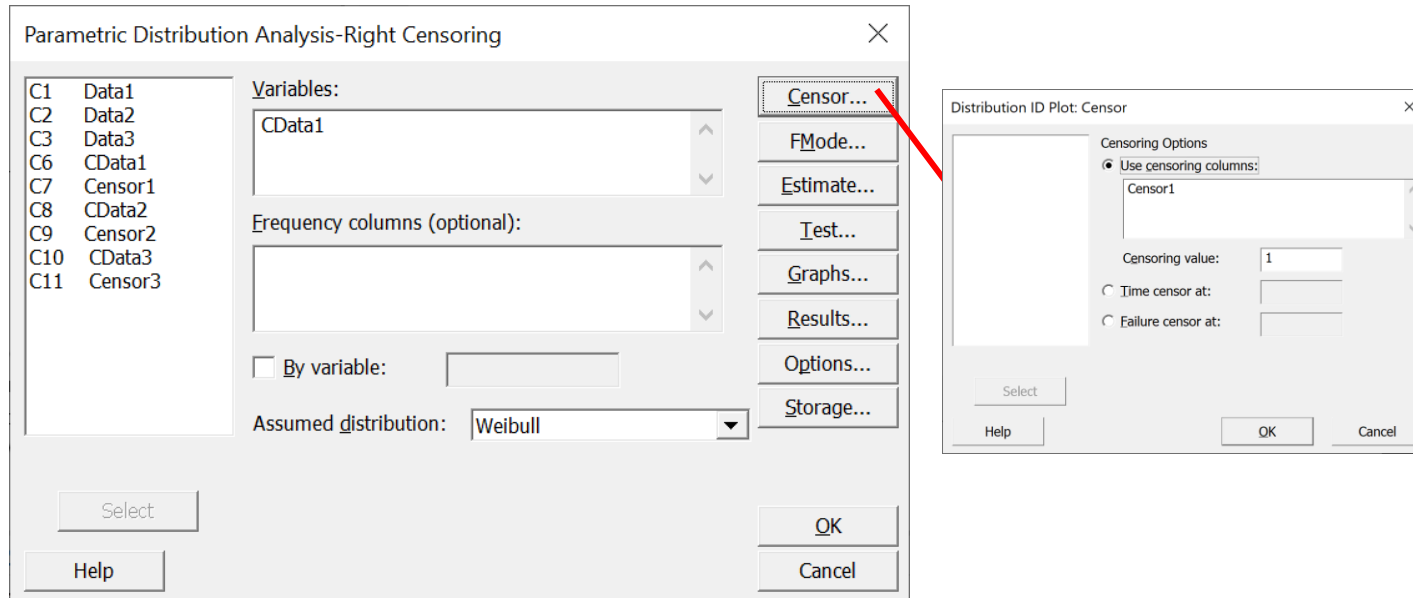
Smaller AD indicates a better fit. Note there isn't much difference with censored data...

Goodness-of-Fit	
Distribution	Anderson-Darling (adj)
Weibull	1130.930
Lognormal	1131.875
Exponential	1134.574
Normal	1132.356

Distribution Fitting – Minitab

With a selected distribution function, fitting in Minitab is in

Stat > Reliability/Survival/Distribution Analysis > Parametric Distribution Analysis



Distribution Fitting – Python

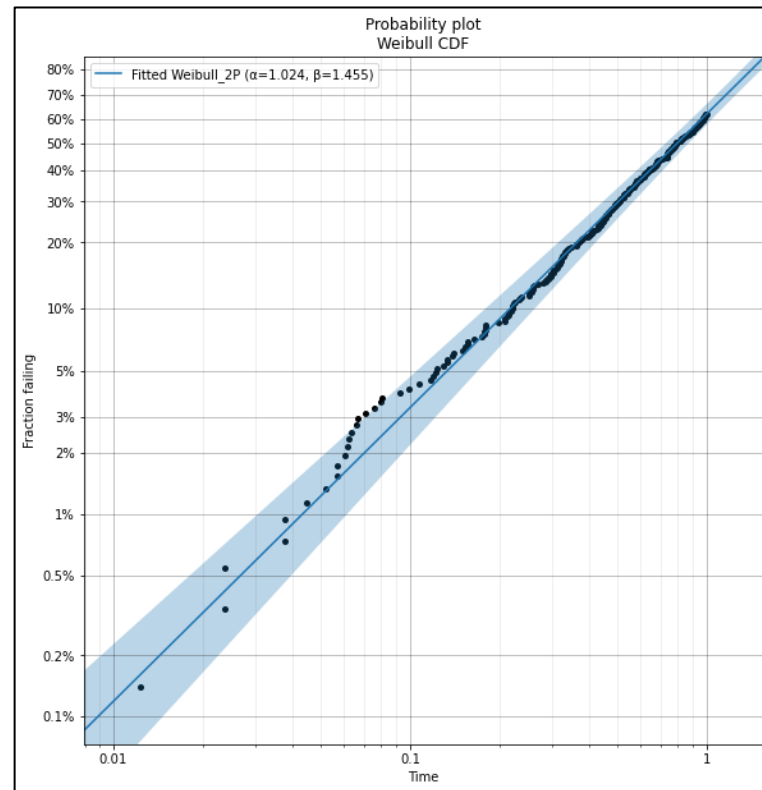
```
import pandas as pd
from reliability.Fitters import Fit_Weibull_2P_grouped

dataDF = pd.read_excel(r'CDData01.xlsx')
wb = Fit_Weibull_2P_grouped(dataframe=dataDF)
```

Results from Fit_Weibull_2P_grouped (95% CI):
 Analysis method: Maximum Likelihood Estimation (MLE)
 Failures / Right censored: 311/189 (37.8% right censored)

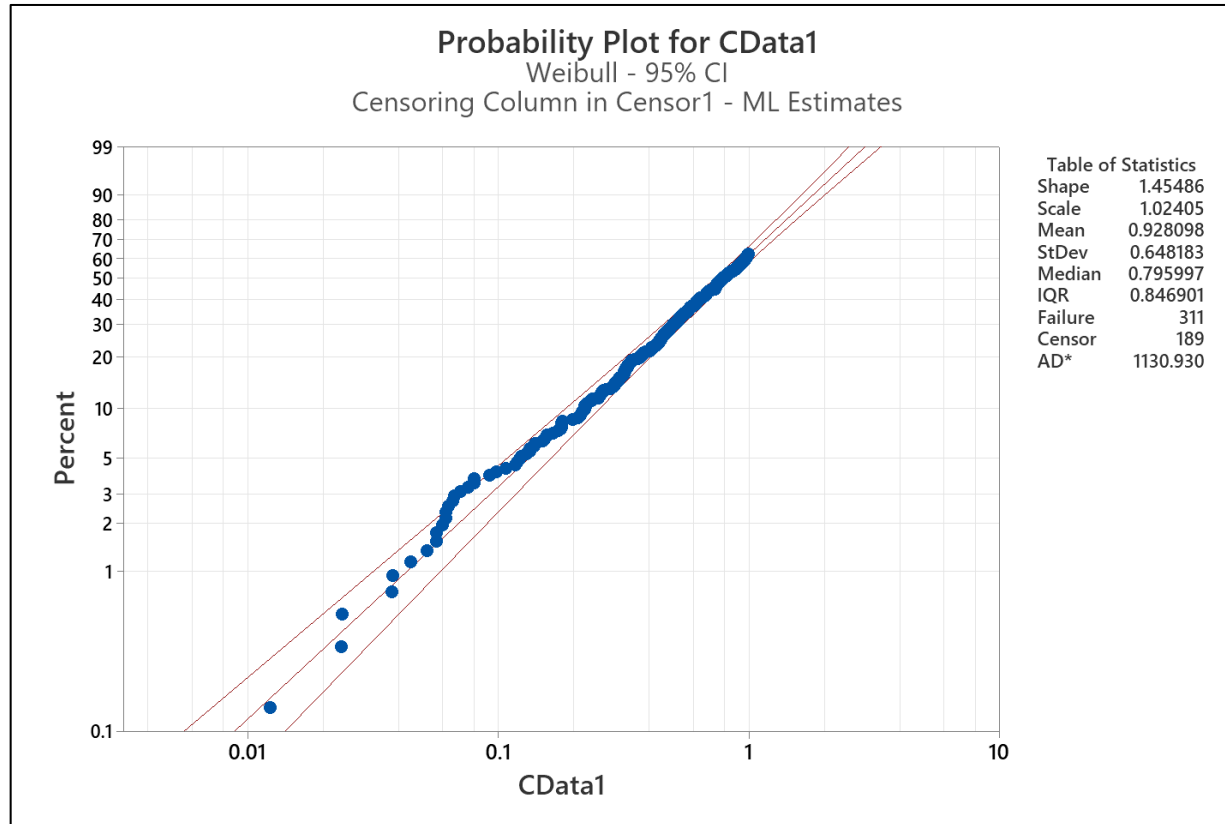
Parameter	Point Estimate	Standard Error	Lower CI	Upper CI
Alpha	1.02405	0.0411941	0.946411	1.10806
Beta	1.45486	0.0746738	1.31562	1.60883

Goodness of fit	Value
Log-likelihood	-326.456
AICc	656.937
BIC	665.342
AD	1130.93



Distribution Fit

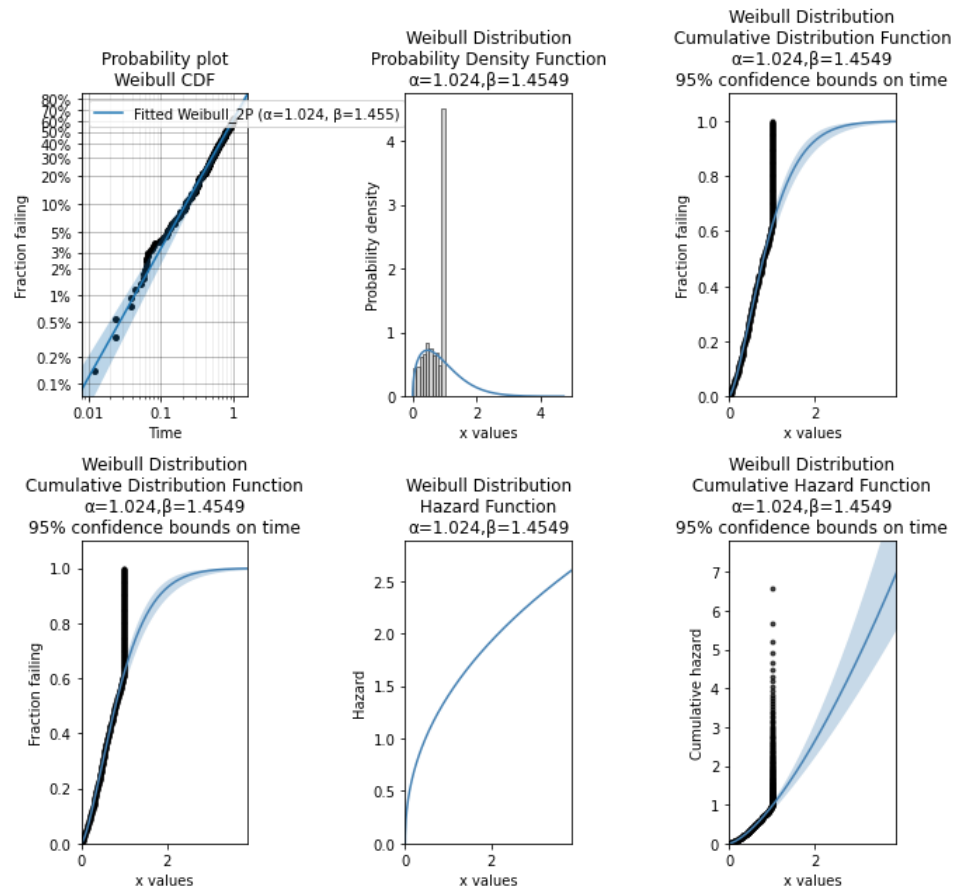
The distribution fit has wider error bars with censored data.



Censored Fitting

Plotting the data on the reliability functions, you can see what censoring does.

You fit the early failures to the function out to the censor time, and match the remaining area under the curve to the fraction not failed.



Effect of Censoring

If we compare the uncensored test-to-failure data set with censored data where the test was stopped after one day:

	Duration	alpha	beta	MTTF	1%tile	5%tile	10%tile
CData 1	1.00	1.463	1.020	0.924	0.044	0.134	0.219
Data 1	3.59	1.455	1.024	0.928	0.043	0.133	0.218

Almost 75% less testing!

Almost identical results!

Exercise

Use the data in “L11 Data.xlsx”

Column “CData1”, “CData2” and “CData3”
with censors “Censor1”, “Censor2” and “Censor 3”

These are the same data as before, but with the test data stopped at mid-point.

- Did the same distribution best-fit?
- How much did the parameters change?
- How much did the MTTF, 1%, 5% and 10% life change?
- How much did the confidence intervals of each of these grow?

Distribution ID Plot-Right Censoring

Variables:
CData1

Frequency columns (optional):

☐ By variable:

☒ Use all distributions
☐ Specify

☒ Distribution 1: Weibull
☒ Distribution 2: Lognormal
☒ Distribution 3: Exponential
☒ Distribution 4: Normal

Select

Help

OK

Cancel

Censor...

Options...

Distribution ID Plot: Censor

Censoring Options

☒ Use censoring columns:
Censor1

Censoring value: 1

☐ Time censor at:

☐ Failure censor at:

Select

Help

OK

Goodness-of-Fit

Distribution Anderson-Darling (adj)

Weibull	1130.930
Lognormal	1131.875
Exponential	1134.574
Loglogistic	1131.112
3-Parameter Weibull	1130.952
3-Parameter Lognormal	1131.023
2-Parameter Exponential	1134.145
3-Parameter Loglogistic	1131.026
Smallest Extreme Value	1133.799
Normal	1132.356
Logistic	1132.511



Overview

Identifying best distributions

Fitting distributions to life data

Censored Data fitting

Service and Sales Data fitting



A system had a part or assembly removed, replaced, or repaired.

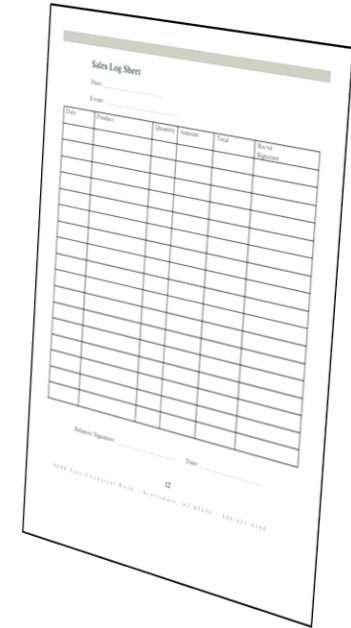
The parts replaced may or may not have been due to your failure mode.

Many sold units may still be in operation, having not failed. Censored values.

[illegible]

Repair Data

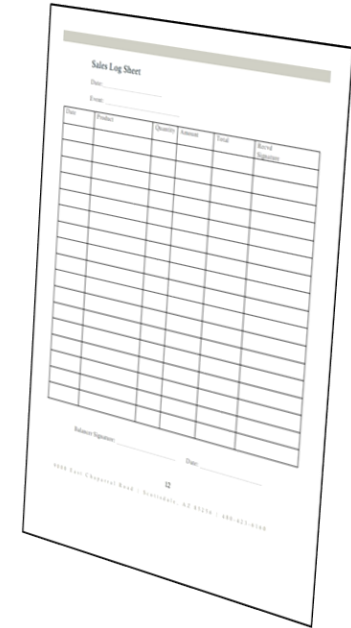
Repair data from the service department only shows the units repaired.
You also need from the sales department how many were sold.
The difference is the units in service not needing repair.



Repair Data

Repair data issues

- Missing values
- You know when a part was repaired.
You don't know when it was last repaired.
You need good data systems for that.
- Binned data
- Multiple failure modes
- Early censored values



Binned Life Data

Often service logs will be repair tallies by time period, say a month.

The data is tallies by months of life, from the month of a sale to the month of the service.

Fit this data to a 2 parameter Weibull distribution.

Months	# Units
5	42
10	62
15	54
20	76
25	79
30	77
35	72
40	76
45	61
50	79
55	65
60	65
65	43
70	67
75	57
80	70
85	48
90	59
95	44
100	55
105	47
110	46
115	34
120	46
125	36
130	44
135	37
140	32
145	29
150	29

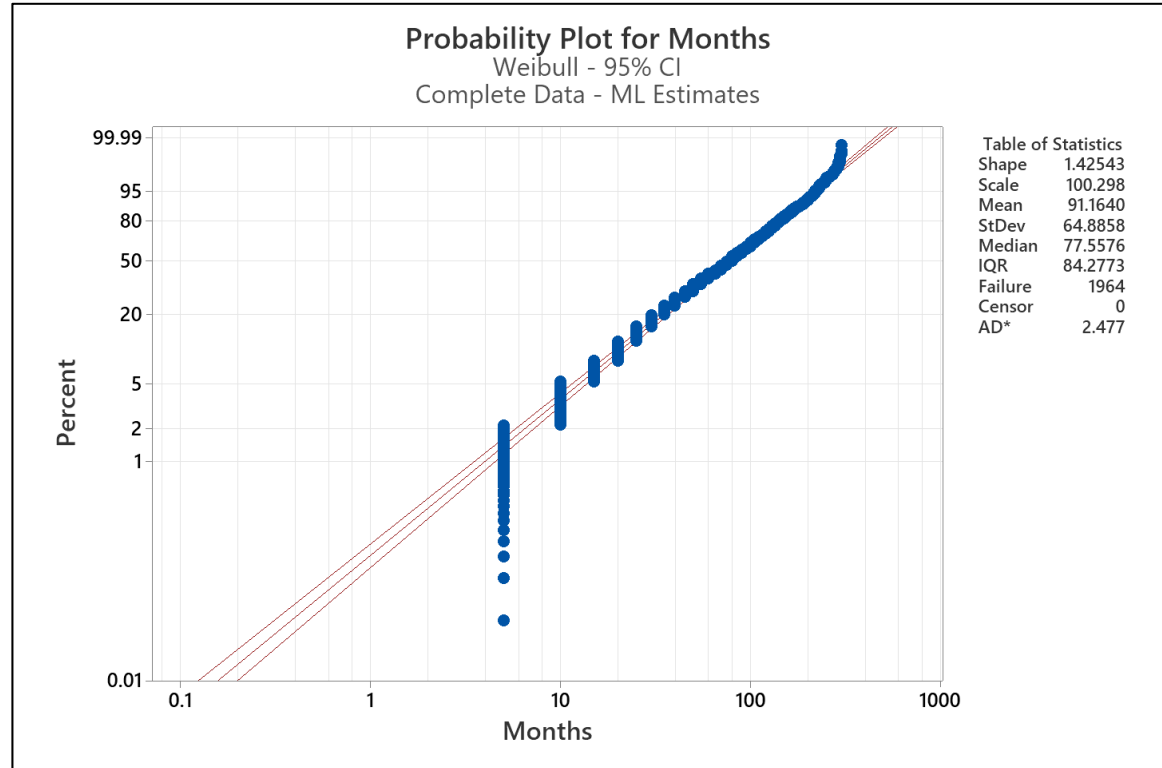
Months	# Units
155	25
160	24
165	25
170	21
175	14
180	12
185	11
190	17
195	17
200	18
205	13
210	14
215	12
220	18
225	9
230	17
235	7
240	4
245	8
250	10
255	2
260	4
265	3
270	4
275	5
280	2
285	4
290	5
295	5
300	3

Binned Life Data

Here the early fit looks disturbing.
It's not. It's what you get
with binned weighted data.

*The 95% confidence intervals
are for the fit line, they are not
prediction intervals for the data.*

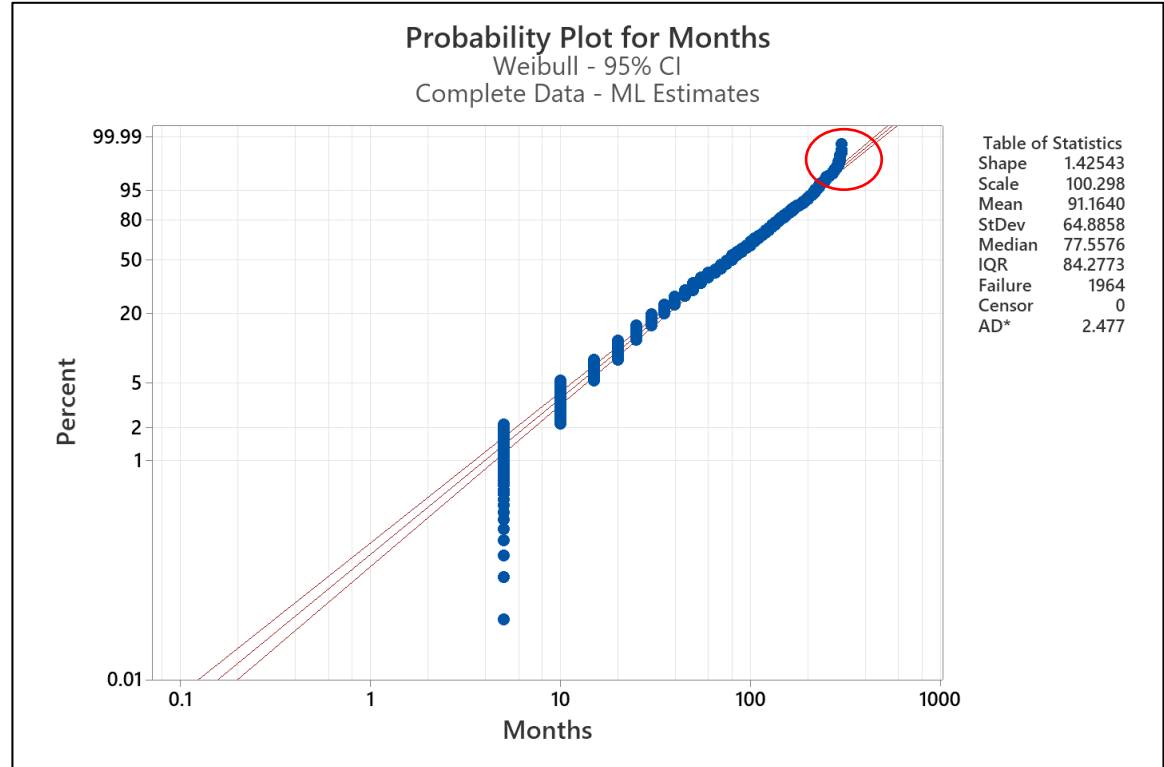
Ensure that the center of the bin
data stays on the percentile line.



Binned Life Data

Here the early fit looks disturbing.
It's not. It's what you get
with binned weighted data.

However, there is lack of fit
at the end of life of the
data. A new different
failure mode is entering
here, atypical of what
happened from months
7-250.

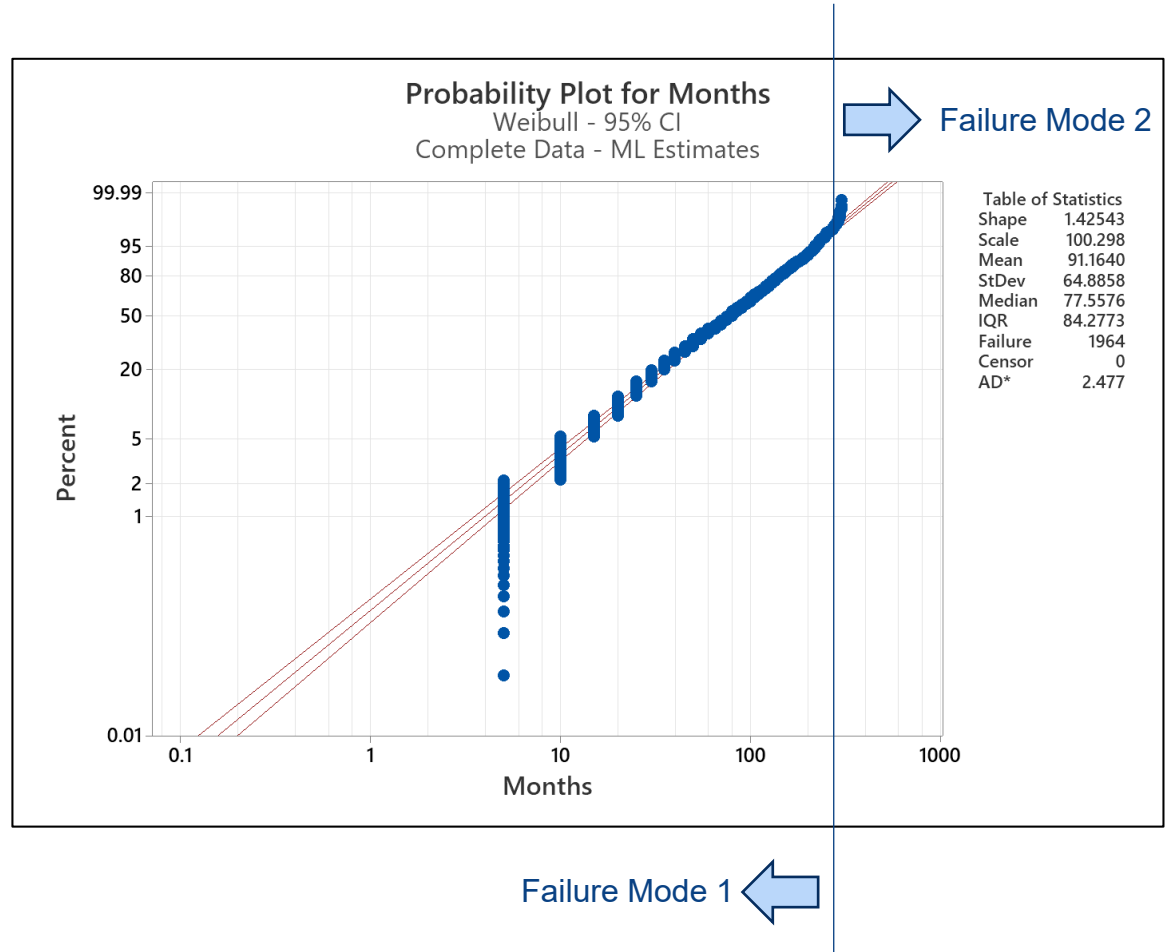


Binned Life Data

Here the early fit looks disturbing.
It's not. It's what you get
with binned weighted data.

Split the data!

A new different failure mode
is entering above 250 months,
atypical of what happened
from months 7-250.



Censor Data of a Different Failure Mode

All values above 250 months don't fit.
They represent a different failure mode.

Censor them.

Add a column “Censor”, with ‘0’
for rows with life < 250, and with ‘1’
for rows with life > 250.

Reanalyze using the censor.

This will analyze only the first data, though
accounting for the fact that the latter did not
fail from this first failure mode and kept going. (Do not just delete them).

Months	Count	Censor
5	42	0
10	62	0
15	54	0
20	76	0
:	:	:
240	4	0
245	8	0
250	10	0
255	2	1
260	4	1
265	3	1
:	:	:
300	3	1

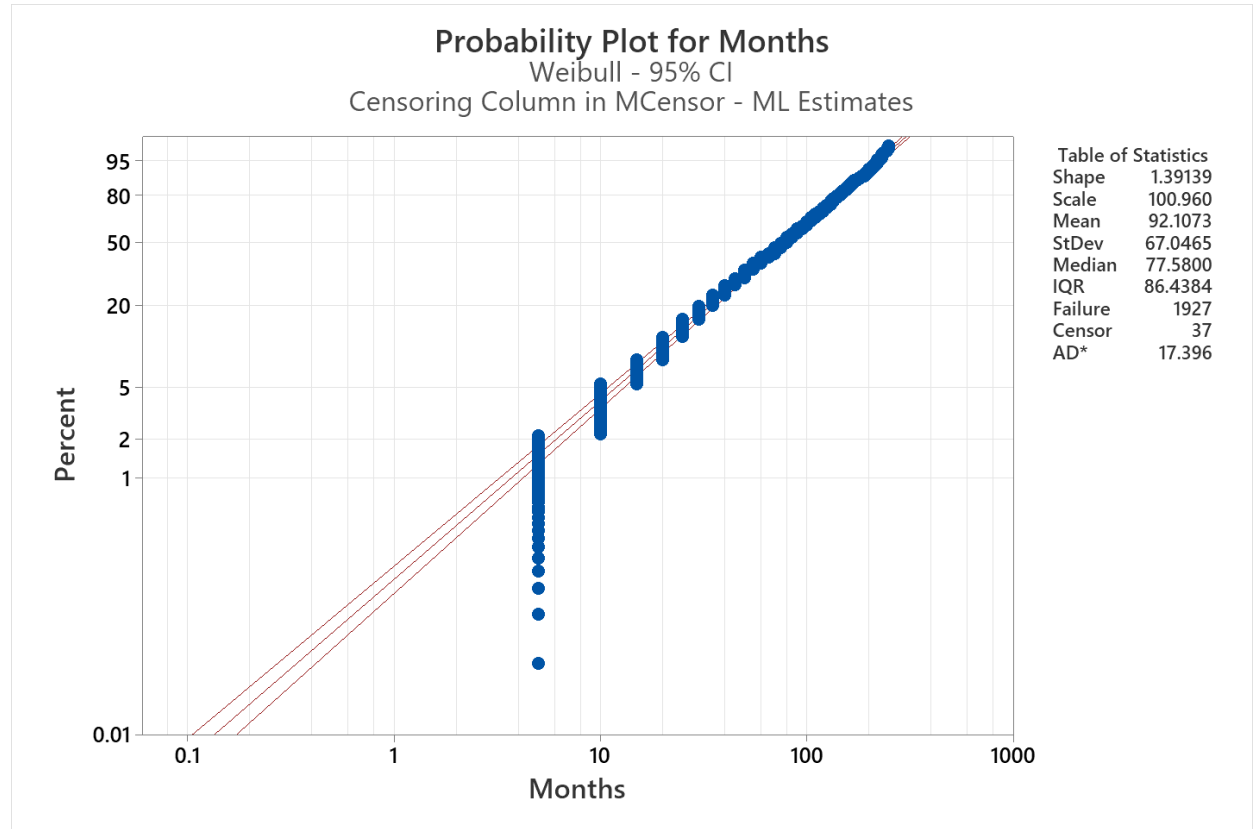
Binned Life Data

Censoring all values above 250 months now shows a good fit.

What kind of failure mode is this?

What is the MTTF?

What is the 1%, 5% life?



Putting it all together...

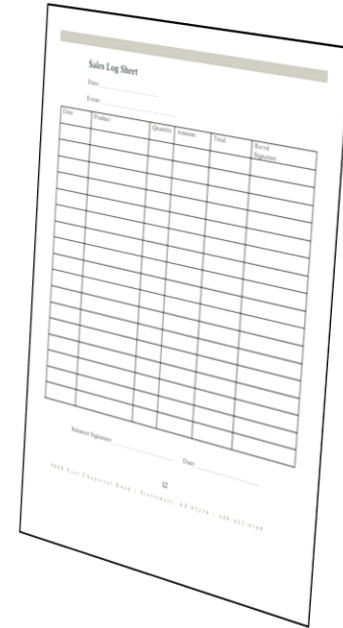
The service logs say what part was repaired and why.

This is helpful.

But reliability analysis needs even better data.

Typically there will be part replacements for failure modes other than yours.

Typically there will be also be remaining units still operating in the field.





Industrial Life Data

Consider the following typical life data for an element in a system that is in the field along with a population of other systems, and the systems get serviced.

You wish to fit the data for a particular failure mode / component.

Failed?	Years	Count
F	1	208
C	1	867
F	2	62
C	2	362
F	3	58
C	3	348
F	4	62
C	4	403
F	5	75
C	5	385
F	6	102
C	6	452
F	7	111
C	7	523
F	8	105
C	8	632
F	9	110
C	9	548
F	10	98
C	10	509
F	11	114
C	11	498
F	12	121
C	12	603
F	13	135
C	13	558
F	14	123
C	14	605
F	15	152
C	15	17432

Industrial Life Data

Consider the following typical life data for a component.

You wish to fit the data for a particular failure mode / component.

The data will include a duration life column
and how many units lasted that long.

Failed?	Years	Count
F	1	208
C	1	867
F	2	62
C	2	362
F	3	58
C	3	348
F	4	62
C	4	403
F	5	75
C	5	385
F	6	102
C	6	452
F	7	111
C	7	523
F	8	105
C	8	632
F	9	110
C	9	548
F	10	98
C	10	509
F	11	114
C	11	498
F	12	121
C	12	603
F	13	135
C	13	558
F	14	123
C	14	605
F	15	152
C	15	17432

Industrial Life Data

Consider the following typical life data for a component.

You wish to fit the data for a particular failure mode / component.

The data will include a duration life column
and how many units lasted that long.

Some parts were replaced for reasons not this failure mode.

So there can be censored values in all life durations.

Failed?	Years	Count
F	1	208
C	1	867
F	2	62
C	2	362
F	3	58
C	3	348
F	4	62
C	4	403
F	5	75
C	5	385
F	6	102
C	6	452
F	7	111
C	7	523
F	8	105
C	8	632
F	9	110
C	9	548
F	10	98
C	10	509
F	11	114
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F	12	121
C	12	603
F	13	135
C	13	558
F	14	123
C	14	605
F	15	152
C	15	17432

Industrial Life Data

Consider the following typical life data for a component.

You wish to fit the data for a particular failure mode / component.

The data will include a duration life column
and how many units lasted that long.

Some parts were replaced for reasons not this failure mode.

So there can be censored values in all life durations.

Finally, there will be a vast majority never serviced.

Here 66% are still in service after 15 years.

Failed?	Years	Count
F	1	208
C	1	867
F	2	62
C	2	362
F	3	58
C	3	348
F	4	62
C	4	403
F	5	75
C	5	385
F	6	102
C	6	452
F	7	111
C	7	523
F	8	105
C	8	632
F	9	110
C	9	548
F	10	98
C	10	509
F	11	114
C	11	498
F	12	121
C	12	603
F	13	135
C	13	558
F	14	123
C	14	605
F	15	152
C	15	17432



Industrial Life Data

Modern reliability analysis software can analyze this most general form of data with no problem.

A column of life.

A column of counts of units that lasted that long.

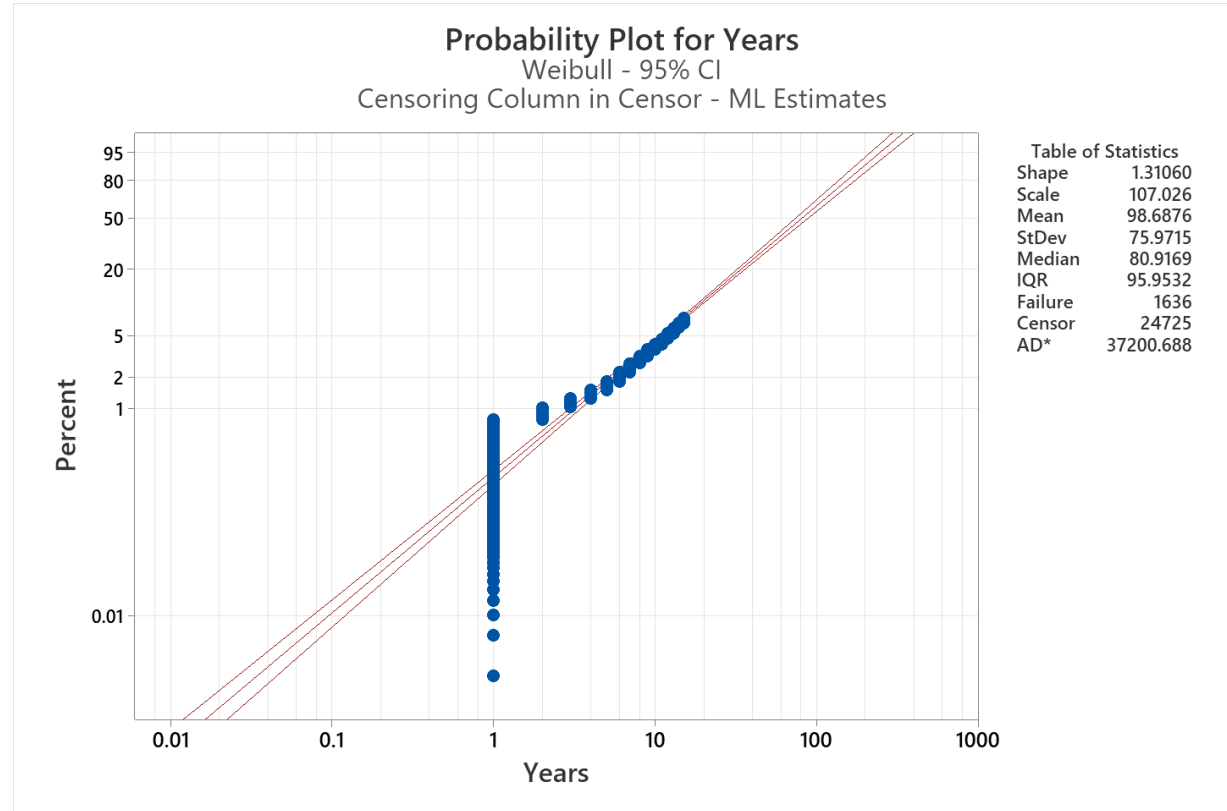
A column of failed or censored (failed for other reasons or not).

Failed?	Years	Count
F	1	208
C	1	867
F	2	62
C	2	362
F	3	58
C	3	348
F	4	62
C	4	403
F	5	75
C	5	385
F	6	102
C	6	452
F	7	111
C	7	523
F	8	105
C	8	632
F	9	110
C	9	548
F	10	98
C	10	509
F	11	114
C	11	498
F	12	121
C	12	603
F	13	135
C	13	558
F	14	123
C	14	605
F	15	152
C	15	17432

Distribution Identification

Resulting fit from the
2 parameter Weibull.

Does the distribution
fit the data?



Distribution Identification

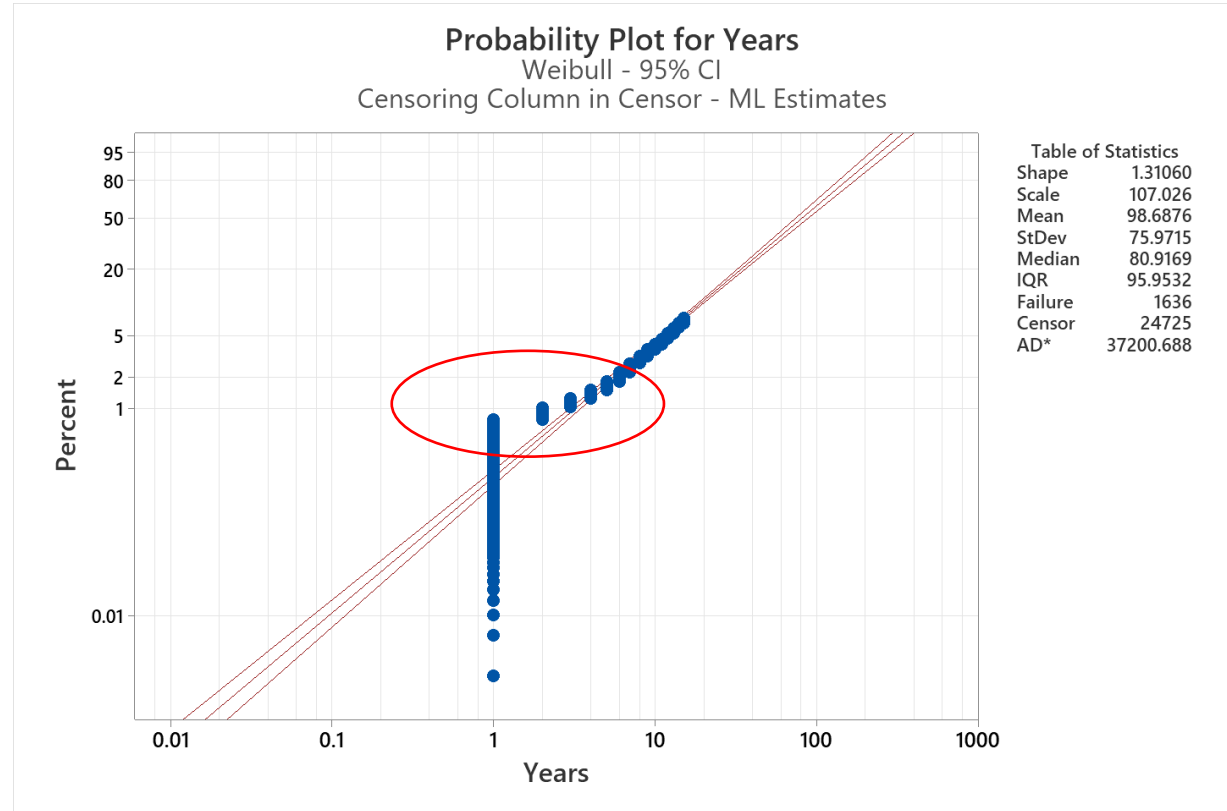
Trend is not fitting.

Year 1 is different.

This is due to the initial
dead-on-arrival units.

A different failure mode
in the early life!

Censor all year 1 data.





Censor Early Life Failure Modes

The first year includes early life manufacturing defective units. Dead on arrival.

Censor these and re-analyze.

Censor	Years	Count
C	1	208
C	1	867
F	2	62
C	2	362
F	3	58
C	3	348
F	4	62
C	4	403
F	5	75
C	5	385
F	6	102
C	6	452
F	7	111
C	7	523
F	8	105
C	8	632
F	9	110
C	9	548
F	10	98
C	10	509
F	11	114
C	11	498
F	12	121
C	12	603
F	13	135
C	13	558
F	14	123
C	14	605
F	15	152
C	15	17432

Weibull Life Fit to Service and Sales Data

Now each year's data
is centered / aligned
with the distribution.

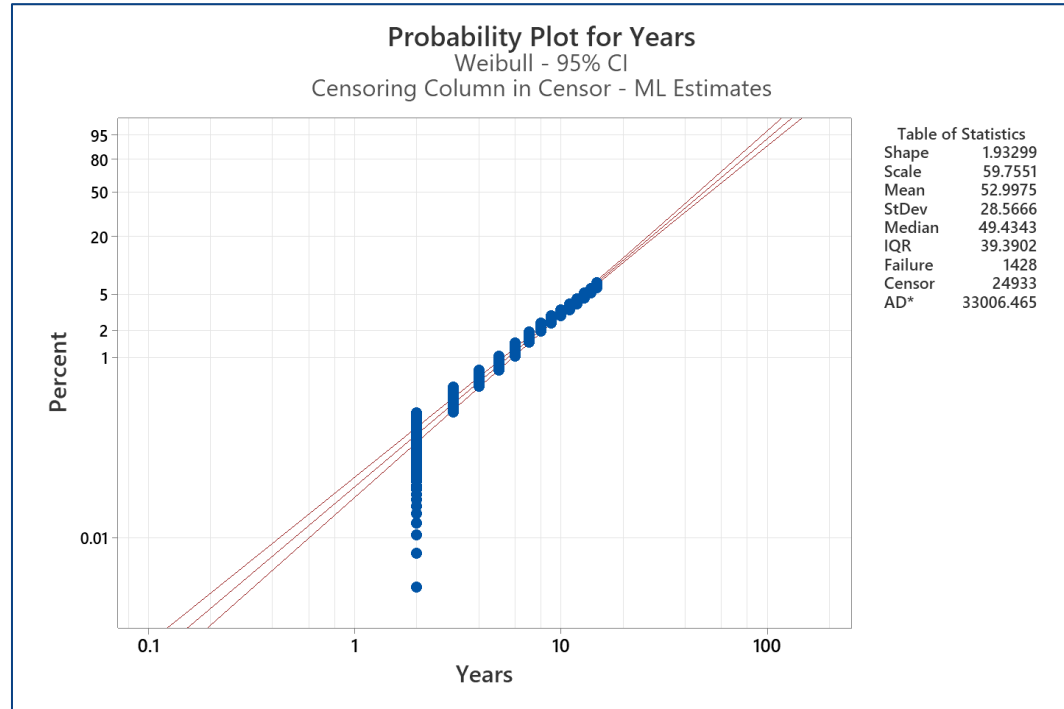
A reasonable fit.

Service data.

Sales data.

Censored.

Weighted.



Weibull Life Fit to Service and Sales Data

Notice the service data ends at 15 years.

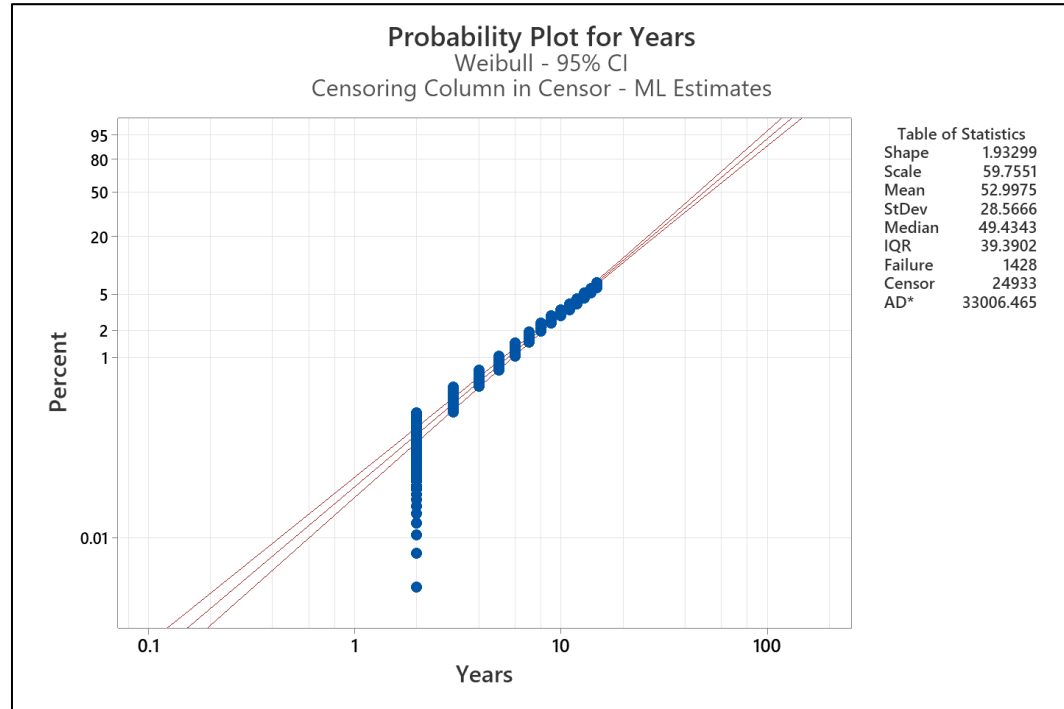
Only the first 34% failed in these first 15 years.

66% are still in service.
When will they fail?

What is the MTTF?

What is the 1% and 5% life?

What type failure mode is this?





Reliability Distributions Summary

Distribution fitting makes use of modern maximum likelihood estimation.

Typical reliability data includes units that did not fail when taken out of service and units that have not yet failed, and also counts of units per time period.

MLE fitting distributions with life, counts, and censoring is all necessary.

Distribution fitting is for a single failure mode. Data must be censored to isolate failure data to the single failure mode.

Reliability Improvement and Testing

Quantifying reliability life takes time.

Each sample unit must be tested to failure.

This can take years for one part!

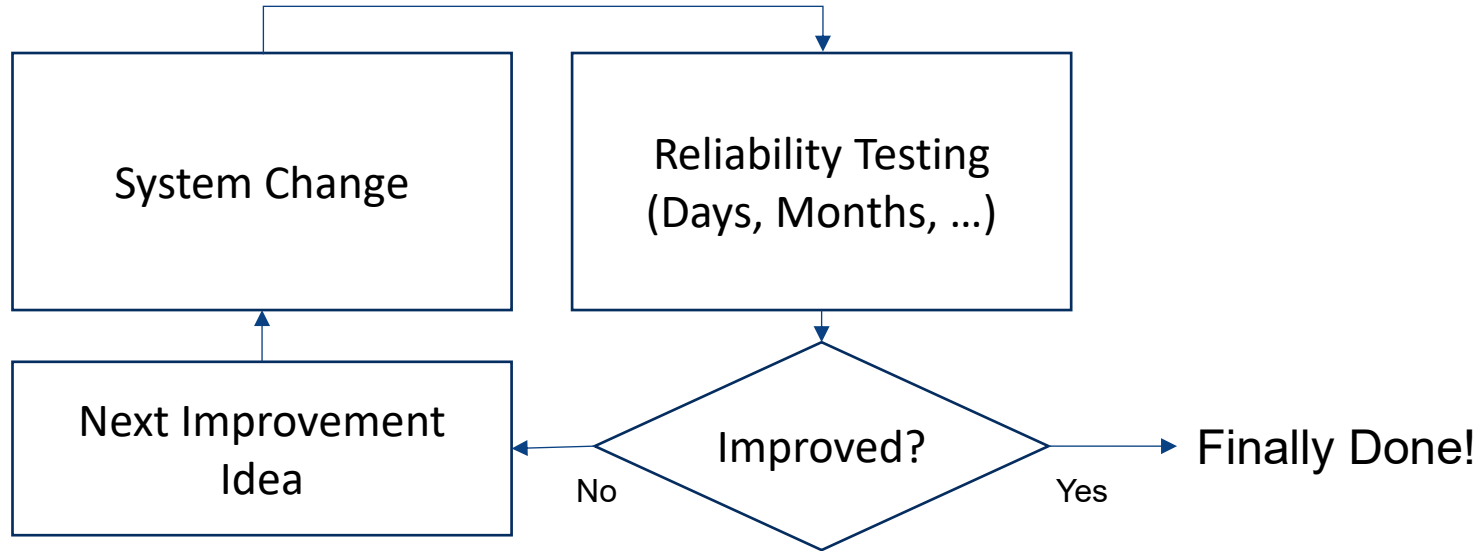
There are advanced methods like ***accelerated life testing*** which can shorten this.

But it remains tedious.



Reliability Improvement

Imagine iteratively testing and improving a system. Often it is impractical.



Reflection: Reliability or Quality Improvement

The material on reliability is important to demonstrating how long a system will operate without failures. What if it shows insufficient reliability?

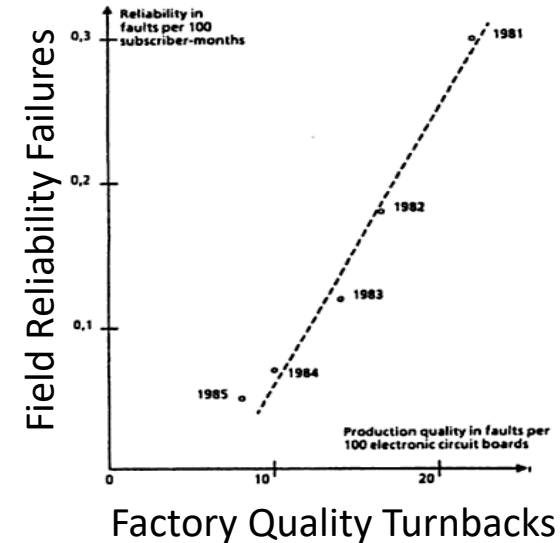
Reliability improvement should not use iterative reliability assessment.

Reliability testing takes too long!

Rather, the reliability problem ought be diagnosed to specific failure modes.

Each individual failure mode should be iteratively improved on Cpk.

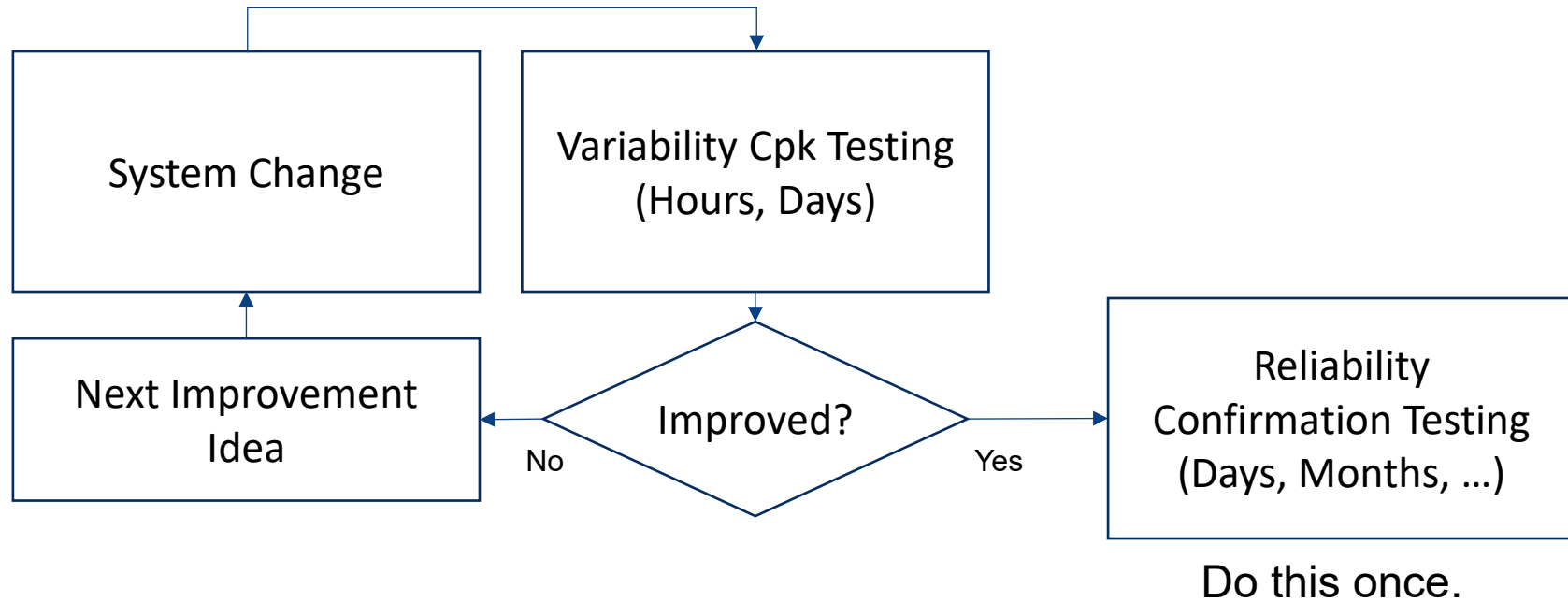
Improve quality, and reliability will follow.



Equipment and System Reliability™, Quality and Reliability Engineering International, Vol.3, pp.195-201, 1987.

Reliability Improvement

Instead improve quality. Reduce variability, which improves reliability.



NEXT TOPIC: SPC and Supplier Quality

EXERCISE – Binned Data Analysis

Use the file “L11 Reliability Data.xlsx”

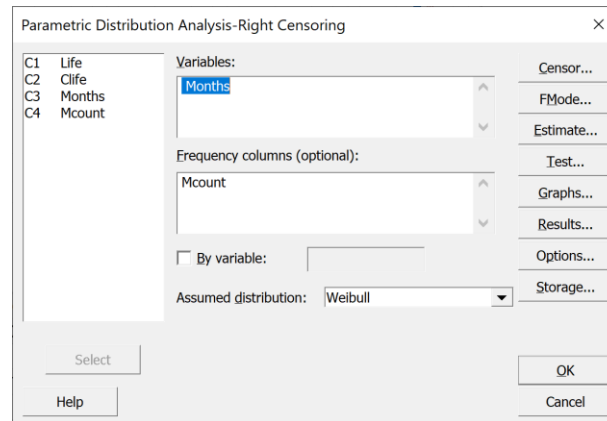
Column “Months” and “MCount” are the months duration and how many units failed at that bin of duration.

Analyze this for the best fit Weibull distribution.

Binned Life Data

The life data is in the “Months” column

The frequency weightings by month are
in the “Mcount” column



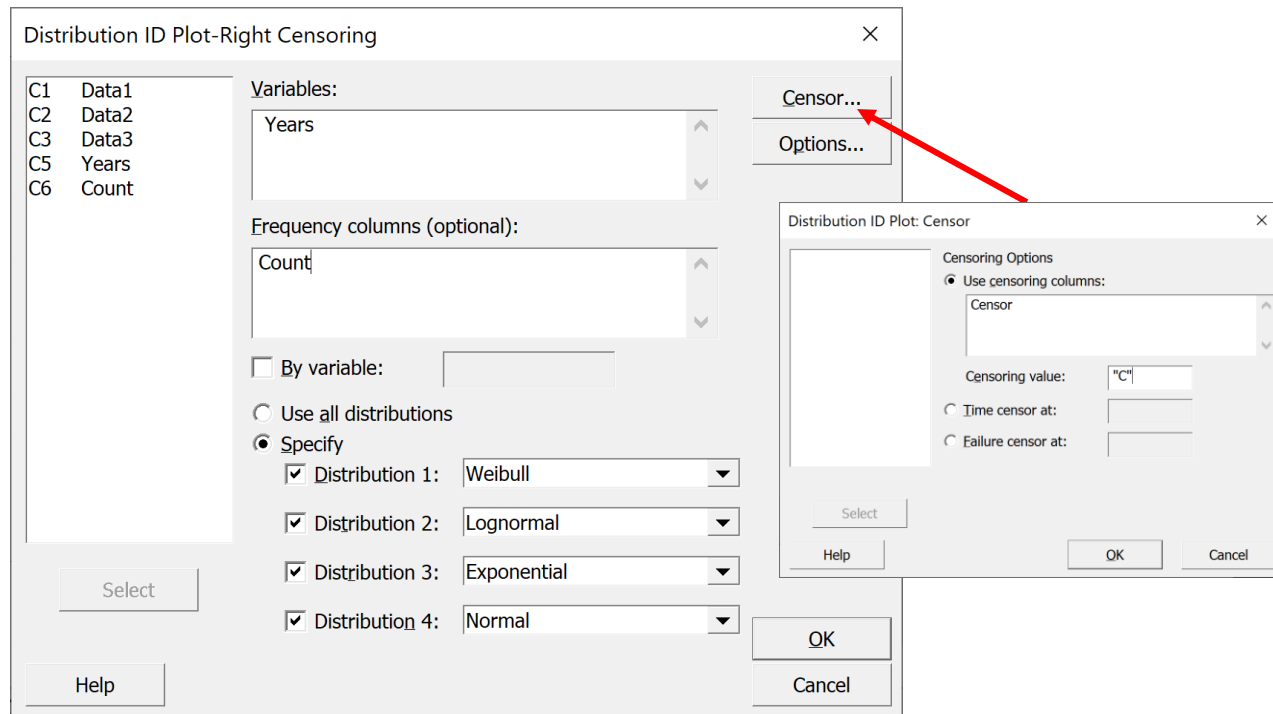
Censored Data Fitting – Minitab

In Minitab fitting is much the same.

Select the life data.

Select the weighting data.

Select the censor data
and censor value.



Censored Data Fitting – Python

In Python create two representations of the data.

One is a Pandas dataframe. The other is an Numpy matrix of failures and censored times.

```
dataDF = pd.read_excel(r'WData01.xlsx')
data = xlsx_to_XCN(path=r'WData01.xlsx')
dataFR = XCN_to_FR(data.X, data.C, data.N)
dataDF = dataDF.rename(columns={"Years": "time", "Censor": "category", "Count": "quantity"})
Fit_Everything(failures=dataFR.failures, right_censored=dataFR.right_censored)
```

Distribution	Alpha	Beta	Gamma	Mu	Sigma	Lambda	Log-likelihood	AICc	BIC	AD
Weibull_Mixture							-8714.12	17438.2	17479.1	31765.7
Weibull_3P	69.7968	1.66581		0.9999			-8915.35	17836.7	17861.2	31765.7
Loglogistic_3P	66.8657	1.69073		0.9999			-8917.25	17840.5	17865	31765.7
Gamma_3P	45.4538	1.74218		0.9999			-8918.33	17842.7	17867.2	31765.7
Weibull_2P	59.755	1.93299					-8927.56	17859.1	17875.5	31765.7
Gamma_2P	32.6237	2.09059					-8928.13	17860.3	17876.6	31765.7
Loglogistic_2P	57.5326	1.96257					-8928.37	17860.7	17877.1	31765.7
Weibull_DS	59.7551	1.93299					-8927.56	17861.1	17885.7	31765.7
Lognormal_2P			4.5683	1.23581			-8933.25	17870.5	17886.9	31765.7
Lognormal_3P		0	4.5683	1.23581			-8933.25	17872.5	17897	31765.7
Weibull_CR							-8929.46	17866.9	17899.6	31765.7
Normal_2P			31.4755		11.0605		-9040.56	18085.1	18101.5	31765.7
Exponential_2P			0.9999			0.00468237	-9087.72	18179.4	18195.8	31766
Gumbel_2P			28.0269		4.90413		-9089.9	18183.8	18200.2	31765.8
Exponential_1P						0.00430988	-9206.1	18414.2	18422.4	31766.1